

Estimating the sampling distribution of the maximum likelihood estimator of the parameters of a series system given a sample of masked system failures

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Abstract

First, the *parametric family* of an abstract *series system* is derived. Then, the *asymptotic sampling distribution* (*multivariate normal*) of a *maximum likelihood estimator* of the true parameter is derived conditioned on the *information* given by a random sample of *masked system failure times*. The asymptotic sampling distributions of statistics that are functions of the parameter follow as a result. Finally, these results are applied to series systems in which component lifetimes are *exponentially* distributed, which lead to closed formulas on a set of *minimally sufficient statistics*.

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1 Introduction

Discuss latent or hidden variables.

Draw a graph in prob. model section. What do we know, how we know it, etc.

We consider series systems consisting of some fixed number of components with uncertain lifetimes. We desire a mathematical model of the system that may be used to predict the failure time of the system and its component cause. However, since the system lifetime and the component cause of failure is uncertain, we are interested in probabilistically modeling the system lifetime, e.g., there is a 75% chance that component 3 will cause a system failure in the next 3 years.

We assume the system belongs to some parametric family but whose true parameter index is unknown. Furthermore, we also assume there is a sample of *masked system failure times* that carry information about the true parameter index.

Using the information in the sample, we employ the frequentist approach of repeated experiments which induces a sampling distribution on the maximum likelihood point estimator of the fixed but unknown true parameter index where *a priori* any value in the parameter space is equally likely.

The point estimator converges to a multivariate normal sampling distribution with a mean given by the true parameter index and a variance that is inversely proportional to sample size.

2 Probabilistic model

A probabilistic model is specified by equations involving random variables which make assumptions about how observable data about the system is generated. In what follows, we describe our model and observable data.

2.1 Series system

The system consists of m components. We make the following assumption about the state of the components.

Assumption 1. *The j^{th} component is initially in a non-failed state and permanently enters a failed state at some uncertain time $t_j > 0$ for $j = 1, \dots, m$.*

Components have uncertain lifetimes as given by the following definition.

Definition 2.1. *The lifetime of the j^{th} component is given by the continuous random variable $T_j \in (0, \infty)$ for $j = 1, \dots, m$.*

We make the following assumption about the joint distribution of the component lifetimes.

Assumption 2. *The random variables $T_1, \dots, T_m$ are mutually independent.*

The system is a series system.

Assumption 3. *The system is in a non-failed state if all m components are in a non-failed state, otherwise the system is in a failed state.*

A system in which at least k of the components must be in a non-failed state for the system to be in a non-failed system is denoted a k -out-of- m system, thus a series system is an m -out-of- m system.

The uncertain lifetime of the system is given by the following definition.

Theorem 2.1. *The lifetime of the system is given by the random variable*

$$S = \min(T_1, \dots, T_m) \in (0, \infty), \quad (2.1)$$

where $T_1, \dots, T_m$ are the component lifetimes.

Proof. A series system fails whenever any component fails, therefore its lifetime is equal to the lifetime of the component with the minimum lifetime. $\square$

2.2 Probability distribution functions and parametric families

As random variables, it is not certain which values $S, T_1, \dots, T_m$ will realize. However, their respective probability distribution functions quantify the uncertainty. The cumulative distribution function is given by the following definition.

Definition 2.2. *Let X be some random variable. The probability that $X \leq x$ is given by its cumulative distribution function, denoted by*

$$F_X(x) = P[X \leq x]. \quad (2.2)$$

The probability density function is given by the following definition.

Definition 2.3. *Let X be some continuous random variable. The relative likelihood that $X = x$ is given by its probability density function, denoted by*

$$f_X(x) = \frac{d}{dt} F_X(t). \quad (2.3)$$

By the Second Fundamental Theorem of Calculus, the probability that a continuous random variable $X \in (a, b]$, $a \leq b$, is given by

$$P[a < X \leq b] = F_X(b) - F_X(a) = \int_a^b f_X(s) ds. \quad (2.4)$$

By these definitions, the relative likelihood that the system will fail at time t is $f_S(t)$ and probability the j^{th} component will before time t is $F_{T_j}(t)$.

Notation. *We simplify the notation and let F_j denote F_{T_j} and f_j denote f_{T_j} . Column vectors are denoted by boldface lower case letters, e.g., $\mathbf{x}$, and row vectors are denoted by the transpose of column vectors, e.g., $\mathbf{x}^T$. Matrices are denoted by boldface upper case greek letters, e.g., $\mathbf{A}$. The j^{th} column of $\mathbf{A}$ is denoted by $\mathbf{A}_j$ (or $[\mathbf{A}]_j$) and the (j, k) -th element of $\mathbf{A}$ is denoted by $\mathbf{A}_{jk}$ (or $[\mathbf{A}]_{jk}$).*

We restrict our attention to parametrized families of probability distribution functions. The system lifetime has a probability density function that is a member of the following parametric family.

Definition 2.4. *The system lifetime has a probability density function that is a member of the parametric family of probability density functions denoted by*

$$\mathcal{F}_{\Theta} = \left\{ f_S(\cdot | \Theta) : \Theta \in \mathbb{R}^{q \times m} \right\}, \quad (2.5)$$

where $\Theta_j \in \Omega$ corresponds to the parameter index of the j^{th} component lifetime.

Particular values of Θ index particular probability density functions within the $\mathcal{F}_{\Theta}$ -family. The true probability density function of the system lifetime is given by the following definition.

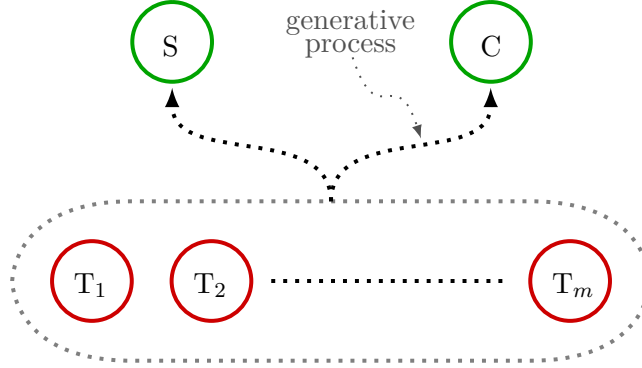
Definition 2.5. *The distribution of the system lifetime has a true parameter index denoted by Θ^* , and we say that*

$$S \sim f_S(\cdot | \Theta^*). \quad (2.6)$$

By these definitions, it follows that

$$T_j \sim f_j(\cdot | \Theta_j^*). \quad (2.7)$$

Figure 1: Graphical model of random variables where $S = \min\{T_1, T_2, \dots, T_m\}$ and C is generated by some process that is a function of Θ^* .



2.3 Masked component failures

By theorem 2.1, if a system failure occurs at time t , then one of the components failed at time t (only a single component can cause a system failure since two or more continuous random variables cannot realize the same value).

Definition 2.6. *The discrete random variable indicating the component responsible for a system failure is denoted by K with a sample space $\mathcal{K} = \{1, \dots, m\}$. That is, $K = j$ indicates that $T_j < T_k$ for $k \in \mathcal{K} \setminus \{j\}$.*

We may not be certain about which component caused a system failure, but we may be able to narrow the component cause to some *subset*. Random candidate failure sets is given by the following definition.

Definition 2.7. *Let C denote the random candidate set with a sample space $2^{\mathcal{K}}$.*

We suppose there is some process that, informed by some degree of knowledge about the components, generates random candidate sets C that *plausibly* failed as depicted in ???. Note that, since our parametric model has a true parameter Θ^* , the candidate set must be some function of Θ^* so that it may offer information about the true parameter index.

Failed candidate sets represent the notion of a diagnostician who is able to, with varying degrees of success, isolate the failed component to some subset of the system. We consider a particular generative model given by the following definition.

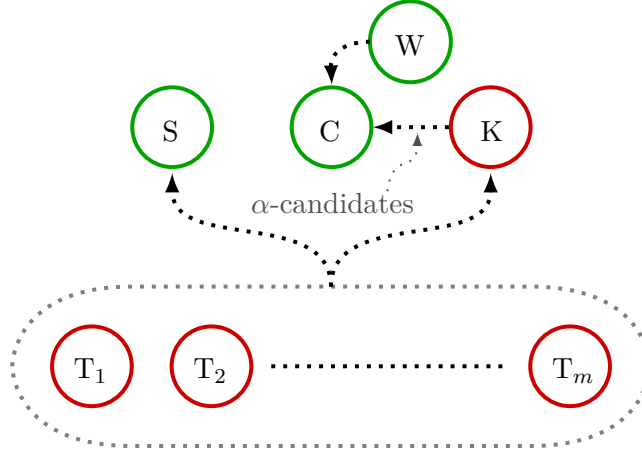
Definition 2.8. *A random α -candidate set is a random candidate set C in which the probability the failed component is included in the set of components is $\alpha \cdot 100\%$.*

The *cardinality* of a component set could theoretically convey information about the system. As a simplification, we make the following assumption.

Assumption 4. *The cardinality of the random set C , denoted by the discrete random variable*

$$W = |C|, \quad (2.8)$$

Figure 2: Graphical model of random variables where $S = \min\{T_1, T_2, \dots, T_m\}$, C is a random α -candidate set, K is the random component failure, and W is the cardinality of random candidate sets.



is independent of the random system lifetime S and the random component cause C and therefore does not carry information about the true parameter index Θ^* .

The indicator function and the k -subset functions are needed for subsequent material and are given by the following definitions.

Definition 2.9 (Indicator function). *Given a subset $\mathcal{A}$ of a set $\mathcal{B}$, the indicator function $\mathbb{1}_{\mathcal{A}}: \mathcal{B} \mapsto \{0, 1\}$ is equal to 1 if $x \in \mathcal{A}$ and 0 otherwise.*

Definition 2.10. *The subsets of $\mathcal{A}$ that are of cardinality k is defined as*

$$[\mathcal{A}]^k := \{ \mathcal{X} \in \mathcal{A} \mid |\mathcal{X}| = k \}, \quad (2.9)$$

which has a cardinality

$$\binom{|\mathcal{A}|}{k}. \quad (2.10)$$

We are primarily interested in the distribution of candidate sets on the condition that they have a specified cardinality.

Definition 2.11. *The random candidate set C conditioned on $W = w$ has a sample space given by*

$$\mathcal{W}_w = \left[2^K \right]^w. \quad (2.11)$$

By assumption 4, the distribution of W is not dependent on the true parameter index Θ^* .

Definition 2.12. *The probability that $W = |C| = w$ is given by the probability mass function $p_W(w)$.*

Any inferences about the parametric model will be *independent* of W except in the sense that conditioned on larger candidate sets generates estimators with greater variance.

If a sample of masked system failures has variable sized candidate sets, we are free to use any parameter-free estimator of the probability mass function of W , e.g., the empirical distribution as described in appendix C.

The *primary* data point is the *masked system failure* as given by the following definition.

Definition 2.13. A masked system failure consists of a system failure time t and a corresponding set of failed candidates C , denoted by $\langle t, C \rangle$.

Assumption 5. A random masked system failure is a jointly distributed random system lifetime S and random failed candidate set C .

Assumption 6. A random sample of n masked system failures, denoted by

$$\mathbf{M}_n = (\langle S_1, C_1 \rangle, \dots, \langle S_n, C_n \rangle) , \quad (2.12)$$

consists of n independent and identically jointly distributed pairs of system failure times and component sets.

Assumption 7. The only information we have about the system and its true parameter index Θ^* is given by observing a random sample of n masked component failures,

$$(\langle S_1 = t_1, C_1 = C_1 \rangle, \dots, \langle S_n = t_n, C_n = C_n \rangle) . \quad (2.13)$$

Given the assumed model, the object of statistical interest is the (unknown) true parameter index Θ^* . Provided an estimate of Θ^* , any characteristic that is a function of Θ^* can be estimated as a result, thus the primary objective is to use the *information* in a sample of n masked component failures to estimate Θ^* with some quantifiable uncertainty.

3 Derivation of parametric probability distribution functions

Chapter 2 described the probabilistic model. In what follows, we derive the parametric functions that are entailed by the model.

3.1 System lifetime

The complement of the cumulative distribution function is the reliability function and is given by the following definition.

Definition 3.1. Let X be some random variable. The probability that $X > x$ is given by its reliability function, denoted by

$$R_X(x) = P[X > x] = 1 - F_X(x) . \quad (3.1)$$

The reliability function of the system lifetime is given by the following theorem.

Theorem 3.1. The system lifetime has a reliability function given by

$$R_S(t | \Theta^*) = \prod_{j=1}^m R_j(t | \Theta_j^*) , \quad (3.2)$$

where R_j is the reliability function of the j^{th} component.

Proof. By definition 3.1, the reliability function is given by

$$R_S(t | \Theta^*) = P[S > t]. \quad (a)$$

By eq. (2.1), $S = \min(T_1, \dots, T_m)$. Performing this substitution yields

$$R_S(t | \Theta^*) = P[\min(T_1, \dots, T_m) > t]. \quad (b)$$

For the minimum to be larger than t , every component must be larger than t , leading to the equivalent equation

$$R_S(t | \Theta^*) = P[T_1 > t, \dots, T_m > t]. \quad (c)$$

By assumption 2, the component lifetimes are independent, thus by the axioms of probability

$$R_S(t | \Theta^*) = P[T_1 > t] \cdots P[T_m > t]. \quad (d)$$

By ??, the probability that $T_j > t$ is equivalent to the reliability function of the j^{th} component lifetime evaluated at t . Performing these substitutions results in

$$R_S(t | \Theta^*) = \prod_{j=1}^m R_j(t | \Theta_j^*). \quad (e)$$

□

The probability density function of the system's lifetime is given by the following theorem.

Theorem 3.2. *The lifetime of the m -out-of- m system has a probability density function given by*

$$f_S(t | \Theta^*) = \sum_{j=1}^m \left(f_j(t | \Theta_j^*) \prod_{\substack{k=1 \\ k \neq j}}^m R_k(t | \Theta_k^*) \right), \quad (3.3)$$

where R_k and f_k are respectively the reliability and probability density functions of the k^{th} component.

Proof. By ??,

$$f_S(t | \Theta^*) = -\frac{d}{dt} R_S(t | \Theta^*). \quad (a)$$

By theorem 3.1, this is equivalent to

$$f_S(t | \Theta^*) = -\frac{d}{dt} \prod_{j=1}^m R_j(t | \Theta_j^*). \quad (b)$$

By the product rule, this is equivalent to

$$\begin{aligned} f_S(t) &= \frac{d}{dt} R_1(t | \Theta_1^*) \prod_{j=2}^m R_j(t | \Theta_j^*) \\ &\quad - R_1(t | \Theta_1^*) \frac{d}{dt} \prod_{j=2}^m R_j(t | \Theta_j^*). \end{aligned} \quad (c)$$

By ??, we can substitute $-\frac{d}{dt}R_1(t|\Theta_1^*)$ with $f_1(t|\Theta_1^*)$, resulting in

$$\begin{aligned} f_S(t) &= f_1(t) \prod_{j=2}^m R_j(t|\Theta_j^*) \\ &\quad - R_1(t|\Theta_1^*) \frac{d}{dt} \prod_{j=2}^m R_j(t|\Theta_j^*) . \end{aligned} \quad (d)$$

Applying the product rule again results in

$$\begin{aligned} f_S(t|\Theta^*) &= f_1(t|\Theta_1^*) \prod_{j=2}^m R_j(t|\Theta_j^*) \\ &\quad + f_2(t|\Theta_2^*) \prod_{\substack{j=1 \\ j \neq 2}}^m R_j(t|\Theta_j^*) \\ &\quad - R_1(t|\Theta_1^*) R_2(t|\Theta_2^*) \frac{d}{dt} \prod_{j=3}^m R_j(t|\Theta_j^*) . \end{aligned} \quad (e)$$

We see a pattern emerge. Applying the product rule $m-1$ times results in

$$\begin{aligned} f_S(t|\Theta^*) &= \sum_{j=1}^{m-1} f_j(t|\Theta_j^*) \prod_{\substack{k=1 \\ k \neq j}}^m R_k(t|\Theta_k^*) \\ &\quad - \prod_{j=1}^{m-1} R_j(t|\Theta_j^*) \frac{d}{dt} R_m(t|\Theta_m^*) \\ &= \sum_{j=1}^m f_j(t|\Theta_j^*) \prod_{\substack{k=1 \\ k \neq j}}^m R_k(t|\Theta_k^*) . \end{aligned} \quad (f)$$

□

The failure rate function simplifies some of the subsequent material and is given by the following definition.

Definition 3.2. *A random variable X with a probability density function f_X and reliability function R_X has a failure rate function given by*

$$h_X(t) = \frac{f_X(t)}{R_X(t)} . \quad (3.4)$$

The failure rate function of the system lifetime is given by the following theorem.

Theorem 3.3. *The system lifetime has a failure rate function given by*

$$h_S(t|\Theta^*) = \sum_{j=1}^m h_j(t|\Theta_j^*) \quad (3.5)$$

where h_j is the failure rate function of the j^{th} component lifetime.

Proof. By eq. (3.4), the failure rate function for the system is given by

$$h_S(t | \Theta^*) = \frac{f_S(t | \Theta^*)}{R_S(t | \Theta^*)}. \quad (a)$$

Plugging in these parametric functions results in

$$h_S(t | \Theta^*) = \frac{\sum_{j=1}^m \left(f_j(t | \Theta_j^*) \prod_{\substack{k=1 \\ k \neq j}}^m R_k(t | \Theta_k^*) \right)}{\prod_{j=1}^m R_j(t | \Theta_j^*)}, \quad (b)$$

which can be simplified to

$$h_S(t | \Theta^*) = \sum_{j=1}^m f_j(t | \Theta_j^*) / R_j(t | \Theta_j^*) \quad (c)$$

$$= \sum_{j=1}^m h_j(t | \Theta_j^*). \quad (d)$$

□

3.2 Component cause of system failure

According to definition 2.6, the component responsible for a system failure is given by the discrete random variable K . The joint likelihood that component k is the cause of a system failure occurring at time t is given by the following theorem.

Theorem 3.4. *The joint probability density function of random variable K and random system lifetime S is given by*

$$f_{K,S}(k, t | \Theta^*) = f_k(t | \Theta^*) \prod_{\substack{j=1 \\ j \neq k}}^m R_j(t | \Theta_j^*), \quad (3.6)$$

where f_k and R_k are respectively the probability density and reliability functions of the k^{th} component.

Proof. See appendix A.1 for a detailed proof. However, a quick justification follows. The likelihood that component k is the cause of a system failure at time t is equivalent to the likelihood that component k fails at time t and the other components do not fail before time t . According to assumption 2, the random variables $T_1, \dots, T_m$ are mutually independent, therefore the likelihood is the product of the probability density function $f_k(t | \Theta_k^*)$ and the reliability functions $R_j(t | \Theta_j^*)$ for $j \in \mathcal{K} \setminus \{k\}$. □

By definition 2.6, if a system failure occurs at time t , then each component has a particular probability of being the cause as given by the following theorem.

Theorem 3.5. *The conditional probability mass function $p_{K|S}(k | t, \Theta^*)$ maps each component $k \in \mathcal{K}$ to the probability that component k is the cause of the system failure at time t and is given by*

$$p_{K|S}(k | t, \Theta^*) = \frac{h_k(t | \Theta_k^*)}{h_S(t | \Theta^*)}, \quad (3.7)$$

where h_j and h_S are respectively the failure rate functions of the j^{th} component and the system.

Proof. By the axioms of probability,

$$p_{K|S}(k|t, \Theta^*) = \frac{f_{K,S}(k, t | \Theta^*)}{f_S(t | \Theta^*)}, \quad (\text{a})$$

thus we wish to show that

$$\frac{h_k(t | \Theta_k^*)}{h_S(t | \Theta^*)} = \frac{f_{K,S}(k, t | \Theta^*)}{f_S(t | \Theta^*)}. \quad (\text{b})$$

By theorem 3.3, the system's failure rate function is given by

$$h_S(t | \Theta^*) = \sum_{j=1}^m h_j(t | \Theta_j^*). \quad (\text{3.5 revisited})$$

Performing this substitution results in

$$\frac{h_k(t | \Theta_k^*)}{h_S(t | \Theta^*)} = \frac{h_k(t | \Theta_k^*)}{\sum_{j=1}^m h_j(t | \Theta_j^*)}. \quad (\text{c})$$

By definition 3.2, the failure rate function of the p^{th} component is given by

$$h_p(t | \Theta_p^*) = \frac{f_p(t | \Theta_p^*)}{R_p(t | \Theta_p^*)}. \quad (\text{d})$$

Performing this substitution results in

$$\frac{h_k(t | \Theta_k^*)}{h_S(t | \Theta^*)} = \frac{f_k(t | \Theta_k^*)}{R_k(t | \Theta_k^*)} \cdot \left[\underbrace{\frac{f_1(t | \Theta_1^*)}{R_1(t | \Theta_1^*)} + \dots + \frac{f_m(t | \Theta_m^*)}{R_m(t | \Theta_m^*)}}_A \right]^{-1}. \quad (\text{e})$$

To combine the fractions labeled A in the above equation, we make their respective denominators the same, resulting in

$$\frac{h_k(t | \Theta_k^*)}{h_S(t | \Theta^*)} = \frac{f_k(t | \Theta_k^*)}{\underbrace{R_k(t | \Theta_k^*)}_B} \cdot \frac{\overbrace{\prod_{j=1}^m R_j(t | \Theta_j^*)}^C}{\sum_{j=1}^m \left[\prod_{\substack{p=1 \\ p \neq j}}^m R_p(t | \Theta_p^*) \right] f_j(t | \Theta_j^*)}. \quad (\text{f})$$

Dividing the part of the above equation labeled C in the numerator by the part labeled B in the denominator results in

$$\frac{h_k(t | \Theta_k^*)}{h_S(t | \Theta^*)} = \frac{f_k(t | \Theta_k^*) \prod_{\substack{j=1 \\ j \neq k}}^m R_j(t | \Theta_j^*)}{\sum_{j=1}^m f_j(t | \Theta_j^*) \prod_{\substack{p=1 \\ p \neq j}}^m R_p(t | \Theta_p^*)}. \quad (\text{g})$$

By eq. (3.3), the denominator in the above equation is the density $f_S(t | \Theta^*)$ and, by eq. (3.6), the numerator is the joint density $f_{K,S}(k, t | \Theta^*)$. Performing these substitutions results in

$$\frac{h_k(t | \Theta_k^*)}{h_S(t | \Theta^*)} = \frac{f_{K,S}(k, t | \Theta^*)}{f_S(t | \Theta^*)}. \quad (\text{h})$$

□

Corollary 3.5.1. *The joint probability density function of random variable K and random system lifetime S is given by*

$$f_{K,S}(k, t | \Theta^*) = h_k(t | \Theta_k^*) R_S(t | \Theta^*) . \quad (3.8)$$

The proof of eq. (3.8) immediately follows from eqs. (3.6) and (3.7).

3.3 Candidate α -sets

The series system consists of m components with random lifetimes $T_1, \dots, T_m$. These random variables are *unobservable* and thus, for instance, we cannot know with certainty from a sample of masked system failures which component failed.

The α -candidate sets have a conditional probability distribution given by the following theorem.

Theorem 3.6. *The random α -candidate set C conditioned on $W = w$ and $K = k$ has a probability distribution given by*

$$f_{C|K,W}(C | k, w, \alpha) = \frac{\alpha}{\binom{m-1}{w-1}} \mathbb{1}_C(k) + \frac{1-\alpha}{\binom{m-1}{w}} \mathbb{1}_{\bar{C}}(k) \quad (3.9)$$

with a sample space $\mathcal{W}_w$.

Proof. Suppose we condition on $K = k$, i.e., the k^{th} -component failed. By ??, a random α -candidate set C contains k with probability α and does *not* contain k with probability $1 - \alpha$.

We partition the sample space $\mathcal{W}_w := [\mathcal{K}]^w$ into the set of sets containing k and its complement. There are ${}^{m-1}C_{w-1}$ candidate sets that contain k since we may place k in the “first” position and of the remaining $w - 1$ positions uniformly choosing any of the remaining $m - 1$ components. There are ${}^{m-1}C_w$ candidate sets that do not contain k since, by removing k from the set of m choices, uniformly choosing w of the remaining $m - 1$ components.

By definition, the outcomes in the partition containing k occur with probability α and we assign uniform probability $({}^{m-1}C_{w-1})^{-1}$ to any such outcome. By the product rule the joint probability that a random α -candidate set contains k is

$$\frac{\alpha}{\binom{m-1}{w-1}} . \quad (a)$$

Similarly, the outcomes in the partition not containing k occur with probability $1 - \alpha$ and we assign uniform probability $({}^{m-1}C_w)^{-1}$ to any such outcome. By the product rule the joint probability that a random α -candidate set does not contain k is

$$\frac{1-\alpha}{\binom{m-1}{w}} . \quad (b)$$

□

The joint probability density function of C and S is given by the following theorem.

Theorem 3.7.

$$f_{C,S|W}(C, t | w, \alpha, \Theta^*) = \sum_{k=1}^m f_{C,K,S|W}(C, k, t | w, \alpha) \quad (3.10)$$

with a sample space $\mathcal{W}_w \times (0, \infty)$.

Proof. The joint distribution is given by

$$f_{C,K,S|W}(\mathcal{C}, k, t | w, \alpha) = f_{C|K,W}(\mathcal{C} | k, w, \alpha) f_{K,S}(k, t | \Theta^*). \quad (a)$$

However, given only a masked failure set, K is *unobservable*, and so we *marginalize* over K . $\square$

By assumption 4, the random lifetime S and the cardinality of the random candidate set $W = |\mathcal{C}|$ are independent, thus by the axioms of probability the joint density of C , S , and W is given by

$$f_{C,S,W}(\mathcal{C}, t, w | \Theta^*) = p_{C|S,W}(\mathcal{C} | t, w, \Theta^*) p_W(w) f_S(t | \Theta^*). \quad (3.11)$$

If it is given that $W = w$, by the axioms of probability

$$f_{C,S|W}(\mathcal{C}, t | w, \Theta^*) = p_{C|S,W}(\mathcal{C} | t, w, \Theta^*) f_S(t | \Theta^*). \quad (3.12)$$

The *optimal* mask is provided by an $\alpha = 1.0$ masked candidate set as given by the following corollary.

Corollary 3.7.1. *The joint probability density of C and S conditioned on $W = w$ and $\alpha = 1.0$ is given by*

$$f_{C,S|W}(\mathcal{C}, t | w, \alpha = 1.0, \Theta^*) = \frac{R_S(t | \Theta^*)}{\binom{m-1}{w-1}} \sum_{k \in \mathcal{C}} h_k(t | \Theta^*) \quad (3.13)$$

with a sample space $\mathcal{W}_w \times (0, \infty)$.

Corollary 3.7.2. *The conditional probability mass function $p_{C|S,W}(\mathcal{C} | t, w, \Theta^*)$ maps each candidate set $\mathcal{C}$ to the probability that the given candidate set contains the component responsible for the system failure at time t . By the axioms of probability,*

$$p_{C|S,W}(\mathcal{C} | t, w, \Theta^*) = \frac{\sum_{j \in \mathcal{C}} h_j(t | \Theta_j^*)}{h_S(t | \Theta^*)} \mathbb{1}_{\mathcal{W}_w}(\mathcal{C}), \quad (3.14)$$

where h_S and h_j are the failure rate functions of the system and the j^{th} component respectively.

Proof. By the axioms of probability,

$$p_{C|S,W}(\mathcal{C} | t, w, \Theta^*) = \frac{f_{C,S|W}(\mathcal{C}, t | w, \Theta^*)}{f_S(t | \Theta^*)}, \quad (a)$$

where $f_{C,S|W}(\cdot | \Theta^*)$ is the joint probability density function given by corollary 3.7.1. Making this substitution immediately yields the result. $\square$

4 Likelihood and Fisher information

In what follows, we frequently prefer to work with vectors rather than matrices.

Definition 4.1. Let $\text{vec}: \mathbb{R}^{m \times q} \mapsto \mathbb{R}^{m \cdot q}$ denote the linear transformation of a matrix of m rows and q columns into a column vector of $m \cdot q$ elements, where consecutive matrix rows are placed side by side. Additionally, the j^{th} component of a vector $\mathbf{a}$ is denoted by $[\mathbf{a}]_j$ or a_j .

Notation. By definition 4.1, Θ^* has an equivalent representation denoted by $\theta^* \equiv \text{vec}(\Theta^*)$ and the j^{th} component of θ^* is denoted by θ_j^* . We use matrix and vector representations interchangeably.

The likelihood of observing a particular realization, $\mathbf{M}(w, n) = \mathbf{m}_n$, is given by the following definition.

Definition 4.2. *A random sample of n masked system failure times conditioned on candidate sets of cardinality w has a joint probability density given by*

$$f_{\mathbf{M}(w,n)}(\mathbf{m}_n | w, \Theta^*) = \prod_{i=1}^n f_S(t_i | \Theta^*) \cdot p_{C|S,W}(C_i | t_i, w, \Theta^*), \quad (4.1)$$

where $f_S(\cdot | \Theta^*)$ is the marginal density given by theorem 3.2 and $p_{C|S,W}(\cdot | t, w, \Theta^*)$ is the conditional probability mass given by corollary 3.7.2.

The generative model of $\mathbf{M}(w, n)$, which directly follows from eq. (4.1), is given by algorithm 1.

Algorithm 1: Generative model of a masked system failure time with w candidates.

Result: a realization of a masked system failure time with w candidates.

Input:

Θ^* , the true parameter value.

w , the cardinality of the candidate sets.

Output:

$\mathbf{m}$, a realization of C, S conditioned on $W = w$.

```

1 Model generate_msft( $w, \Theta^*$ )
2   draw a system lifetime  $t \sim f_S(\cdot | \Theta)$ 
3   draw a candidate set  $\mathcal{C} \sim p_{C|S,W}(\cdot | t, w, \Theta)$ 
4    $\mathbf{m} \leftarrow \langle t, \mathcal{C} \rangle$ 
5   return  $\mathbf{m}$ 

```

The subsequent material makes a few assumptions, denoted *regularity conditions*, given by the following.

Assumption 8. *The following regularity conditions hold for the $\mathcal{F}_\Theta$ -family of the series system:*

1. θ^* is interior to the parameter support Ω .
2. The log-likelihood ℓ is continuous, thrice differentiable, and bounded.
3. The true parameter value θ^* is identified, i.e.,

$$\theta^* = \arg \max_{\theta \in \Omega} \mathbb{E}_{\theta^*} \left(\ln f_{C,S|W}(C, S | w, \theta) \right). \quad (4.2)$$

By definition 4.2, the probability density that $\mathbf{M}(w, n) = \mathbf{m}_n$ is given by

$$f_{\mathbf{M}(w,n)}(\mathbf{m}_n | w, \theta^*) = \prod_{i=1}^n f_{C,S|W}(C_i, t_i | w, \theta^*). \quad (4.1 \text{ revisited})$$

If we fix $\mathbf{m}_n$ and allow the parameter θ to change, we have the likelihood function.

Definition 4.3. *The likelihood function represents the likelihood of parameter index $\boldsymbol{\theta}$ conditioned on $\mathbf{M}(w, n) = \mathbf{m}_n$ and is given by*

$$\mathcal{L}(\boldsymbol{\theta} | w, \mathbf{m}_n) = \prod_{i=1}^n f_{C,S|W}(C_i, t_i | w, \boldsymbol{\theta}). \quad (4.3)$$

The likelihood is a measure of the extent to which the given sample provides support for particular values of a parameter in $\mathcal{F}_{\boldsymbol{\theta}}$; its surface captures all the information there is about $\boldsymbol{\theta}^*$ from the given sample.

The log-likelihood plays a more central role than the likelihood for reasons that will be made clear later on.

Definition 4.4. *The log-likelihood with respect to $\boldsymbol{\theta}$ given a particular $\mathbf{M}(w, n) = \mathbf{m}_n$ is given by*

$$\ell(\boldsymbol{\theta} | w, \mathbf{m}_n) = \sum_{i=1}^n \ln f_{C,S|W}(C_i, t_i | w, \boldsymbol{\theta}). \quad (4.4)$$

In subsequent material, we need the gradient operator given by the following definition.

Definition 4.5 (Gradient operator). *The gradient operator $\nabla: (\mathbb{R}^q \mapsto \mathbb{R}) \mapsto (\mathbb{R}^q \mapsto \mathbb{R}^q)$ is defined as*

$$\nabla g(x_1, \dots, x_q) = \left(\frac{\partial g(x_1, \dots, x_q)}{\partial x_1}, \dots, \frac{\partial g(x_1, \dots, x_q)}{\partial x_q} \right)^T. \quad (4.5)$$

The score function is given by the following definition.

Definition 4.6 (Score function). *The Fisher's score function, denoted by $s: \mathbb{R}^{m \cdot q} \mapsto \mathbb{R}^{m \cdot q}$, is the gradient of the log-likelihood function*

$$s(\boldsymbol{\theta} | w, \mathbf{m}_n) := \nabla \ell(\boldsymbol{\theta} | w, \mathbf{m}_n). \quad (4.6)$$

The variance-covariance is given by the following definition.

Definition 4.7 (Variance-covariance). *A random vector $\mathbf{X} \in \mathbb{R}^p$ has a variance-covariance given by*

$$\text{Var}(\mathbf{X}) = \mathbb{E} \left[(\mathbf{X} - \mathbb{E}[\mathbf{X}]) (\mathbf{X} - \mathbb{E}[\mathbf{X}])^T \right], \quad (4.7)$$

which is a p -by- p semi-positive definite matrix.

The score function when given a particular realization $\mathbf{m}_n$ is a statistic. When we consider it as a function of a random sample, it is a random vector. When evaluated at the true parameter index $\boldsymbol{\theta}^*$, it has a variance-covariance given by the following definition.

Definition 4.8 (Fisher information matrix). *The score as a function of a random sample $\mathbf{M}(w, n)$ has a variance-covariance known as the Fisher information matrix, a $(m \cdot q)$ -by- $(m \cdot q)$ matrix denoted by*

$$\mathcal{I}_n(\boldsymbol{\theta}^* | w) = \text{Var}_{\boldsymbol{\theta}^*} [s(\boldsymbol{\theta}^* | w, \mathbf{M}(w, n))], \quad (4.8)$$

where the variance is taken with respect to the joint probability density function of $C, S | W = w$ indexed by $\boldsymbol{\theta}^$.*

The Hessian is given by the following definition.

Definition 4.9 (Hessian operator). *The Hessian of a function $g: \mathbb{R}^q \mapsto \mathbb{R}$ with respect to $\mathbf{x} \in \mathbb{R}^q$ is a q -by- q matrix where the (j, k) -th element is given by*

$$\mathcal{H}(g(\mathbf{x}))_{jk} = \frac{\partial^2 g(\mathbf{x})}{\partial x_j \partial x_k}. \quad (4.9)$$

Under the regularity conditions, the information matrix may be computed from the Hessian and is given by the following postulate.

Postulate 4.1. *The information matrix is given by the expectation of the negative of the Hessian of the log-likelihood function ℓ evaluated at $\boldsymbol{\theta}^*$. That is,*

$$\mathcal{I}_n(\boldsymbol{\theta}^* | w) = -\mathbb{E}_{\boldsymbol{\theta}^*} \left[\mathcal{H}(\ell(\boldsymbol{\theta} | w, \mathbf{M}(w, n))) \Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}^*} \right], \quad (4.10)$$

where the expectation is taken with respect to $\boldsymbol{\theta}^*$ and the Hessian is taken with respect to $\boldsymbol{\theta}$ and then evaluated at $\boldsymbol{\theta}^*$.

For clarity, the (i, j) -th element of the information matrix is given by the following definition.

Definition 4.10. *The (i, j) -th element of $\mathcal{I}_n(\cdot | w)$ is given by*

$$\mathcal{I}_n(\boldsymbol{\theta}^* | w)_{ij} = - \sum_{\mathcal{C} \in \mathcal{W}_w} \int_0^\infty \left[\frac{\partial^2}{\partial \theta_i \partial \theta_j} \ln f_{\mathcal{C}, S | W}(\mathcal{C}_i, t_i | w, \boldsymbol{\theta}) \right] \Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}^*} \times f_{\mathcal{C}, S | W}(\mathcal{C}_i, t_i | w, \boldsymbol{\theta}^*) dt, \quad (4.11)$$

where $\mathcal{W}_w \times (0, \infty)$ is the sample space.

Notation. We denote the Fisher information matrix of $\boldsymbol{\theta}^*$ given $\mathbf{M}(w, 1)$ by

$$\mathcal{I}(\boldsymbol{\theta}^* | w) \equiv \mathcal{I}_1(\boldsymbol{\theta}^* | w). \quad (4.12)$$

The information matrix is a key quantity in large sample theory; it quantifies the *expected* information (as a reduction of uncertainty) $\mathbf{M}(w, n)$ carries about $\boldsymbol{\theta}^*$. For sufficiently large samples, the log-likelihood has a Taylor series approximation given by

$$\ell(\boldsymbol{\theta} | w) \approx -\frac{1}{2} (\boldsymbol{\theta} - \boldsymbol{\theta}^*)^\top \mathcal{I}_n(\boldsymbol{\theta}^* | w) (\boldsymbol{\theta} - \boldsymbol{\theta}^*), \quad (4.13)$$

which is maximized at $\boldsymbol{\theta}^*$. Intuitively, the more “peaked” the log-likelihood is expected to be at $\boldsymbol{\theta}^*$, the more information a random sample is expected to carry about $\boldsymbol{\theta}^*$.

The information independent samples have about $\boldsymbol{\theta}^*$ is *additive* as given by the following postulate.

Postulate 4.2. *Algebraically, the Fisher information independent samples have about the true parameter index $\boldsymbol{\theta}^*$ is additive such that*

$$\mathcal{I}_{n_1+n_2}(\boldsymbol{\theta}^* | \cdot) = \mathcal{I}_{n_1}(\boldsymbol{\theta}^* | \cdot) + \mathcal{I}_{n_2}(\boldsymbol{\theta}^* | \cdot). \quad (4.14)$$

By postulate 4.2, the Fisher information of a random sample of n masked system failure times drawn from $\mathbf{M}(w, n)$ is given by

$$\mathcal{I}_n(\boldsymbol{\theta}^* | w) = n \mathcal{I}(\boldsymbol{\theta}^* | w) . \quad (4.15)$$

A masked system failure time in which the cardinality of the candidate set is distributed according to W , as described by ??, has a Fisher information matrix given by the following theorem.

Theorem 4.1. *The Fisher information matrix of true parameter index $\boldsymbol{\theta}^*$ with respect to a realization of jointly distributed random system lifetime $S = t$ and candidate set $C = \mathcal{C}$ is given by*

$$\mathcal{I}(\boldsymbol{\theta}^*) = \sum_{w=1}^{m-1} p_W(w) \mathcal{I}(\boldsymbol{\theta}^* | w) , \quad (4.16)$$

where $p_W(w)$ is the marginal distribution of $W = |C|$.

Proof. Suppose we have a random sample of n masked system failure times. By ??, the expected number of sample points that have w candidates is given by

$$n \cdot p_W(w) . \quad (a)$$

By the additive property of *Fisher information* given by postulate 4.2, the information matrix is given by

$$\mathcal{I}_n(\boldsymbol{\theta}^*) = \sum_{w=1}^{m-1} n \cdot p_W(w) \mathcal{I}(\boldsymbol{\theta}^* | w) . \quad (b)$$

Setting n to 1 results in

$$\mathcal{I}(\boldsymbol{\theta}^*) = \sum_{w=1}^{m-1} p_W(w) \mathcal{I}(\boldsymbol{\theta}^* | w) . \quad (c)$$

□

Remark. *A model is a simplification of reality. The model denoted by $\mathbf{M}_n$ is an approximation of some (unknown) underlying generative model, e.g., W and S may not truly be independent. Since the expected information matrix is an expectation with respect to the simplified model, the expected information is also a simplified approximation. The observed information matrix, given by*

$$\mathcal{J}_n(\hat{\boldsymbol{\theta}}_n | \mathbf{m}_n) = - \mathcal{H} \left(\ell(\boldsymbol{\theta} | \mathbf{m}_n) \right) \Big|_{\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}_n} , \quad (4.17)$$

is a statistic of a real sample and therefore makes less of a commitment to the simplified model. Generally, the observed information matrix is preferable but since we know (by prescription) the underlying generative model,

$$\mathcal{J}_n(\hat{\boldsymbol{\theta}}_n | \mathbf{M}_n) \xrightarrow{P} \mathcal{I}_n(\boldsymbol{\theta}^*) , \quad (4.18)$$

and so we proceed with the expected information matrix. △

5 Sampling distribution of parameter estimators

By definitions 2.4 and 2.5, the random system lifetime S has a probability density function that is a member of the $\mathcal{F}_\Theta$ -family with a true parameter index $\boldsymbol{\theta}^*$. Let some estimator of $\boldsymbol{\theta}^*$ be a function of

the information in a random sample of n masked system failure times. The estimator is a function of a random sample, thus it has *sampling distribution*, a random vector denoted by $\mathbf{Y}_n$. That is,

$$\mathbf{Y}_n = \psi(\mathbf{M}_n) . \quad (5.1)$$

A particular realization of the estimator is denoted by $\bar{\boldsymbol{\theta}}_n$, i.e., $\mathbf{Y}_n = \bar{\boldsymbol{\theta}}_n$. All else being equal, we prefer estimators which vary only *slightly* from sample to sample with a *central tendency* around $\boldsymbol{\theta}^*$. That is, we prefer unbiased estimators in which each component has small variance.

The sampling distribution of $\bar{\boldsymbol{\theta}}_n$ has a variance-covariance given by $\text{Var}(\mathbf{Y}_n)$ (see eq. (4.7)) and a bias given by the following definition.

Definition 5.1. *The bias of a point estimator $\mathbf{Y}_n = \bar{\boldsymbol{\theta}}_n$ is given by*

$$\text{bias}(\mathbf{Y}_n) = \mathbb{E}[\mathbf{Y}_n] - \boldsymbol{\theta}^* . \quad (5.2)$$

We are often faced with a trade-off between bias and variance. A measure of estimator accuracy that is a function of both the bias and the variance is the mean squared error as given by the following definition.

Definition 5.2 (MSE). *The mean squared error of the sampling distribution of $\bar{\boldsymbol{\theta}}_n$ is given by*

$$\text{MSE}(\mathbf{Y}_n) = \mathbb{E}[(\mathbf{Y}_n - \boldsymbol{\theta}^*)^\top (\mathbf{Y}_n - \boldsymbol{\theta}^*)] \quad (5.3)$$

An equivalent way to compute the mean squared error is given by the following postulate.

Postulate 5.1. *The mean squared error of the sampling distribution of $\bar{\boldsymbol{\theta}}_n$ as given by definition 5.2 is equivalent to*

$$\text{MSE}(\mathbf{Y}_n) = \text{tr}(\text{Var}(\mathbf{Y}_n)) + \text{bias}^2(\mathbf{Y}_n) , \quad (5.4)$$

where $\text{tr}(\mathbf{A})$ computes the sum of the diagonal elements of square matrix $\mathbf{A}$.

Fisher information reduces the uncertainty about $\boldsymbol{\theta}^*$. The *minimum-variance unbiased estimator* (UMVUE) has a lower-bound given by the inverse of the *Fisher information matrix*, denoted the *Cramér-Rao lower-bound*. The minimum variance obtainable from a random sample of n masked system failure times drawn from $\mathbf{M}_n$ is given by

$$\mathbf{CRLB}_n = \frac{1}{n} \mathcal{I}^{-1}(\boldsymbol{\theta}^*) . \quad (5.5)$$

By eq. (5.4), if $\bar{\boldsymbol{\theta}}_n$ is an unbiased estimator of $\boldsymbol{\theta}^*$, then the mean squared error is given by

$$\text{MSE}(\mathbf{Y}_n) = \text{tr}(\text{Var}(\mathbf{Y}_n)) . \quad (5.6)$$

By eqs. (5.5) and (5.6), the mean squared error of any unbiased point estimator of $\boldsymbol{\theta}^*$ in which the only *a priori* information is given by a random sample $\mathbf{M}_n$ has a lower-bound given by the trace of $\mathbf{CRLB}_n$,

$$\text{MSE}(\mathbf{Y}) \geq \text{tr}(\mathbf{CRLB}_n) . \quad (5.7)$$

5.1 Maximum likelihood estimator given w candidates

Suppose the only information about the true parameter index θ^* is given by a sample of n masked system failure times in which only those system failures with w candidates are observed. The maximum likelihood estimator based on this conditional sample is given in the following definition.

Definition 5.3. A parameter index $\hat{\theta}(n|w)$ that maximizes the likelihood of observing $\mathbf{M}(w, n) = \mathbf{m}_n$ (see assumption 6) is a maximum likelihood estimator of θ^* . That is,

$$\hat{\theta}(n|w) = \arg \max_{\theta \in \Omega} \mathcal{L}(\theta | w, \mathbf{m}_n), \quad (5.8)$$

where Ω is the feasible parameter space of the parametric family.

As further justification of assumption 5, the maximum likelihood estimator maximizes the likelihood of generating the given $\mathbf{m}_n$ such that for each system failure time a component in the corresponding candidate set is the cause.

The accuracy of the maximum likelihood estimator is a function of the agreement between the model and reality. In our model, the primary point of disagreement is given by the assumption of a series system under a parametric model of independent component lifetimes, i.e., $\mathcal{F}_\Theta$. However, if the model is a reasonable abstraction of the objects of interest,

The distribution of candidate sets, i.e, C , disagrees with the way the actual candidate sets are generated. Even if this is the case, the logic that components that are more likely to have caused a system failure *with respect to the parameters* (other ways in which a component might fail are outside the scope of the parametric model)

The accuracy of the maximum likelihood estimator is a function of the agreement between the model and reality. In our model, the primary point of disagreement is given by the assumption of a series system under a parametric model of independent component lifetimes, i.e., $\mathcal{F}_\Theta$.

Another point of disagreement is the assumption that candidates are generated from the joint probability densities of the component lifetimes $T_1, \dots, T_m$ and random candidate sets C . However, recall the assumption is that, under the assumed model $S = \min(T_1, \dots, T_m)$, components that are more likely to have failed at observable time $S = t$ are more likely to be in the observable candidate set.

Note that *any other* distribution of candidate sets generates a smaller likelihood given the parametric model. We claim this is the *optimal* candidate set since it provides the most information about θ . If the parametric family $\mathcal{F}_\Theta$ is a very accurate model of the reality, then whether the failed component is in the candidate set or not is irrelevant. What matters more is that the most likely components are in the candidate set, as that will provide the most information about θ .

If this is not a very accurate model, the maximum likelihood estimator may not be a consistent estimator even if . Rather, the maximum likelihood estimator will generate an estimate that makes the provided sample the most likely to be generated by the assumed model, i.e., the component lifetimes will not be accurately modeled since they must be modeled in such a way as to make the biased sample more likely.

The logarithm is a monotonically increasing function, thus the parameter value $\hat{\theta}(n|w)$ that maximizes $\mathcal{L}$ also maximizes the log-likelihood ℓ (see definition 4.4), i.e.,

$$\hat{\theta}(n|w) = \arg \max_{\theta \in \Omega} \ell(\theta | w, \mathbf{m}_n). \quad (5.9)$$

According to Bickel [1], if the parameter support Ω of the parametric family is open, the log-likelihood ℓ is differentiable with respect to θ , and $\hat{\theta}(n|w)$ exists, then $\hat{\theta}(n|w)$ must be a stationary

point of ℓ as given by

$$s(\hat{\boldsymbol{\theta}}(n|w) | w, \mathbf{m}_n) = \mathbf{0}, \quad (5.10)$$

where s is the score function given by eq. (4.6). In cases where there are no closed-form solutions to eq. (5.10), iterative methods may be used as described in appendix B.

An estimator of $\boldsymbol{\theta}^*$ given a sample $\mathbf{M}(w, n) = \mathbf{m}_n$ is the maximum likelihood estimator $\hat{\boldsymbol{\theta}}(n|w)$ described in definition 5.3. Since $\hat{\boldsymbol{\theta}}(n|w)$ is a function of the random sample $\mathbf{M}(w, n)$, it has a *sampling distribution*.

Definition 5.4. The sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$ is a random vector denoted by $\mathbf{Y}(n|w) \in \mathbb{R}^{m \cdot q}$.

The generative model for the maximum likelihood estimator, `generate_mle`($w | \cdot$), is described by algorithm 2. This generative model depends on `generate_msft`, the generative model for the *masked system failure time* as described by algorithm 1. Note that in algorithm 2, line 7 may be approximated with `find_mle`, a function that numerically solves the stationary points of the maximum likelihood equations as described by algorithm 5.

Algorithm 2: Generative model of the maximum likelihood estimator conditioned on w candidates

Result: a realization of a maximum likelihood estimate from the sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$.

Input:

- $\boldsymbol{\theta}^*$, the true parameter index.
- w , the cardinality of the candidate set.
- n , the number of masked system failure times.

Output:

$\hat{\boldsymbol{\theta}}(n|w)$, a realization of $\mathbf{Y}(n|w)$.

```

1 Model generate_mle( $n, w, \boldsymbol{\theta}^*$ )
2    $\mathbf{m}_n \leftarrow \emptyset$ 
3   for  $i \leftarrow 1$  to  $n$  do
4      $\mathbf{M} \leftarrow \text{generate\_msft}(w, \boldsymbol{\theta}^*)$ 
5      $\mathbf{m}_n \leftarrow \mathbf{m}_n \cup \{\mathbf{M}\}$ 
6   end
7    $\hat{\boldsymbol{\theta}}(n|w) \leftarrow \arg \max_{\boldsymbol{\theta} \in \Omega} \ell(\boldsymbol{\theta} | w, \mathbf{m}_n)$ 
8   return  $\hat{\boldsymbol{\theta}}(n|w)$ 

```

The asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$ is a consistent estimator of $\boldsymbol{\theta}^*$.

Postulate 5.2. As $n \rightarrow \infty$, the sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$ converges in probability to $\boldsymbol{\theta}^*$, written

$$\mathbf{Y}(n|w) \xrightarrow{P} \boldsymbol{\theta}^*, \quad (5.11)$$

since

$$\lim_{n \rightarrow \infty} \mathbb{P}[\text{MSE}(\mathbf{Y}(n|w)) < \epsilon] = 1 \quad (5.12)$$

for every $\epsilon > 0$.

By eqs. (4.7) and (5.11), the variance-covariance of the asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$ is given by

$$\text{Var}[\mathbf{Y}(n|w)] = \mathbb{E}[(\mathbf{Y}(n|w) - \boldsymbol{\theta}^*)(\mathbf{Y}(n|w) - \boldsymbol{\theta}^*)^\top]. \quad (5.13)$$

Postulate 5.3. *The variance-covariance of the asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$ obtains the Cramér-Rao lower-bound for point estimators that are strictly a function of n masked system failure times in which candidate sets are of cardinality w . We denote this variance-covariance by*

$$\boldsymbol{\Sigma}_n(\boldsymbol{\theta}^* | w) \equiv \frac{1}{n} \boldsymbol{\mathcal{I}}^{-1}(\boldsymbol{\theta}^* | w) . \quad (5.14)$$

The asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$ is normally distributed.

Postulate 5.4. *As $n \rightarrow \infty$, the sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$ converges in distribution to a multivariate normal distribution with a mean $\boldsymbol{\theta}^*$ and a variance-covariance $\boldsymbol{\Sigma}_n(\boldsymbol{\theta}^* | w)$, written*

$$\mathbf{Y}(n|w) \xrightarrow{d} \text{MVN}(\boldsymbol{\theta}^*, \boldsymbol{\Sigma}_n(\boldsymbol{\theta}^* | w)) . \quad (5.15)$$

The maximum likelihood estimator $\hat{\boldsymbol{\theta}}(n|w)$ is an asymptotically *efficient* estimator since it obtains the Cramér-Rao lower-bound as given by ???. Thus, for a sufficiently large sample of masked system failure times, the sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$ varies only *slightly* from sample to sample with a *central tendency* around $\boldsymbol{\theta}^*$.

By eqs. (5.5), (5.6) and (5.11) and ??, the asymptotic sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$ has the minimum mean squared error of any unbiased estimator,

$$\text{MSE}(\mathbf{Y}(n|w)) = \frac{1}{n} \text{tr} \left(\boldsymbol{\mathcal{I}}^{-1}(\boldsymbol{\theta}^* | w) \right) = \text{tr}(\text{CRLB}_{n|w}) . \quad (5.16)$$

Remark. *Look up Slutsky's theorem to justify the next paragraph.* $\triangle$

To estimate the sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$, we may assume the sample size is sufficiently large such that the asymptotic distribution becomes a reasonable approximation. Since $\mathbf{Y}(n|w) \xrightarrow{P} \boldsymbol{\theta}^*$ and $\mathbf{Y}(n|w) \xrightarrow{d} \text{MVN}(\boldsymbol{\theta}^*, \boldsymbol{\Sigma}_n(\boldsymbol{\theta}^* | w))$, it follows that

$$\mathbf{Y}(n|w) \xrightarrow{d} \text{MVN}(\hat{\boldsymbol{\theta}}(n|w), \boldsymbol{\Sigma}_n(\hat{\boldsymbol{\theta}}(n|w) | w)) . \quad (5.17)$$

Thus, we can approximate $\boldsymbol{\theta}^*$ and $\boldsymbol{\mathcal{I}}(\boldsymbol{\theta}^* | w)$ and obtain the following result. The proof of this theorem is beyond the scope of this paper.

Theorem 5.1. *For sufficiently large sample size n , the sampling distribution of $\hat{\boldsymbol{\theta}}(n|w)$ is approximately normally distributed with a mean $\hat{\boldsymbol{\theta}}(n|w)$ and a variance-covariance matrix $\boldsymbol{\Sigma}_n(\hat{\boldsymbol{\theta}}(n|w) | w)$, written*

$$\mathbf{Y}(n|w) \overset{\text{approx.}}{\sim} \text{MVN}(\hat{\boldsymbol{\theta}}(n|w), \boldsymbol{\Sigma}_n(\hat{\boldsymbol{\theta}}(n|w) | w)) . \quad (5.18)$$

5.2 Maximum likelihood estimator

Consider an i.i.d. sample of r asymptotically unbiased estimates of $\boldsymbol{\theta}^*$ denoted by $\bar{\boldsymbol{\theta}}^{(i)}$ which have sampling distributions with variance-covariances given respectively by $\boldsymbol{\Sigma}^{(i)}$ for $i = 1, \dots, r$. The maximum likelihood estimator of $\boldsymbol{\theta}^*$ given these point estimates is the inverse-variance weighted mean and is given by

$$\hat{\boldsymbol{\theta}} = \left(\sum_{i=1}^r \mathbf{A}_i \right)^{-1} \left(\sum_{i=1}^r \mathbf{A}_i \bar{\boldsymbol{\theta}}^{(i)} \right) , \quad (5.19)$$

where $\mathbf{A}_i$ is the inverse of $\Sigma^{(i)}$.

Suppose that the estimates are given by the maximum likelihood estimator described in ?? . The maximum likelihood estimator given these estimates has a sampling distribution given by the following definition.

Definition 5.5. Let $\mathbf{M}_n$ be a random sample of n masked system failure times in which n_i realizations have w_i candidates for $i = 1, \dots, r$. The maximum likelihood estimator given by

$$\hat{\boldsymbol{\theta}}_n = \hat{\boldsymbol{\theta}} = \left(\sum_{i=1}^r \mathbf{A}_i \right)^{-1} \left(\sum_{i=1}^r \mathbf{A}_i \bar{\boldsymbol{\theta}}^{(i)} \right), \quad (5.20)$$

has a sampling distribution given by

$$\mathbf{Y}_n = \left(\sum_{i=1}^r \mathbf{A}_i \right)^{-1} \left(\sum_{i=1}^r \mathbf{A}_i \mathbf{Y}^{(i)}(n_i | w_i) \right), \quad (5.21)$$

where $\mathbf{Y}^{(i)}(n_i | w_i)$ is the sampling distribution of $\hat{\boldsymbol{\theta}}(n_i | w_i)$ for $i = 1, \dots, r$ and

$$\mathbf{A}_i = n_i \mathcal{I}(\boldsymbol{\theta}^* | w_i) \quad (5.22)$$

is the information matrix for $\boldsymbol{\theta}^*$ with respect to $\mathbf{M}(w_i, n_i)$.

Theorem 5.2. The random vector $\mathbf{Y}_n$ is an asymptotically unbiased estimator of $\boldsymbol{\theta}^*$.

Proof. The expectation of $\mathbf{Y}_n$ is given by

$$\mathbb{E}[\mathbf{Y}_n] = \mathbb{E} \left[\left(\sum_{i=1}^r \mathbf{A}_i \right)^{-1} \left(\sum_{i=1}^r \mathbf{A}_i \mathbf{Y}^{(i)}(n_i | w_i) \right) \right] \quad (a)$$

$$= \left(\sum_{i=1}^r \mathbf{A}_i \right)^{-1} \left(\sum_{i=1}^r \mathbf{A}_i \mathbb{E}[\mathbf{Y}^{(i)}(n_i | w_i)] \right) \quad (b)$$

$$= \left(\sum_{i=1}^r \mathbf{A}_i \right)^{-1} \left(\sum_{i=1}^r \mathbf{A}_i \right) \boldsymbol{\theta}^* \quad (c)$$

$$= \boldsymbol{\theta}^*. \quad (d)$$

□

Theorem 5.3. The variance-covariance of $\mathbf{Y}_n$ is given by

$$\text{Var}[\mathbf{Y}_n] = \left(\sum_{i=1}^r n_i \mathcal{I}(\boldsymbol{\theta}^* | w_i) \right)^{-1}. \quad (5.23)$$

Proof. Let

$$\mathbf{B} = \left(\sum_{i=1}^r \mathbf{A}_i \right)^{-1}. \quad (a)$$

The variance-covariance of $\mathbf{Y}_n$ is given by

$$\text{Var}[\mathbf{Y}_n] = \text{Var} \left[\mathbf{B} \left(\sum_{i=1}^r \mathbf{A}_i \mathbf{Y}^{(i)}(n_i | w_i) \right) \right] \quad (\text{b})$$

$$= \mathbf{B} \text{Var} \left[\sum_{i=1}^r \mathbf{A}_i \mathbf{Y}^{(i)}(n_i | w_i) \right] \mathbf{B}^\top \quad (\text{c})$$

$$= \mathbf{B} \left(\sum_{i=1}^r \text{Var} [\mathbf{A}_i \mathbf{Y}^{(i)}(n_i | w_i)] \right) \mathbf{B}^\top \quad (\text{d})$$

$$= \mathbf{B} \left(\sum_{i=1}^r \mathbf{A}_i \text{Var} [\mathbf{Y}^{(i)}(n_i | w_i)] \mathbf{A}_i^\top \right) \mathbf{B}^\top. \quad (\text{e})$$

By ??, the variance-covariance of $\mathbf{Y}^{(i)}(n_i | w_i)$ is given by

$$\frac{1}{n_i} \mathcal{I}^{-1}(\boldsymbol{\theta}^* | w_i). \quad (\text{f})$$

By eq. (5.22), this is equivalent to $\mathbf{A}_i^{-1}$. Performing this substitution results in

$$\text{Var}[\mathbf{Y}_n] = \mathbf{B} \left(\sum_{i=1}^r \mathbf{A}_i \mathbf{A}_i^{-1} \mathbf{A}_i^\top \right) \mathbf{B}^\top \quad (\text{g})$$

$$= \mathbf{B} \left(\sum_{i=1}^r \mathbf{A}_i^\top \right) \mathbf{B}^\top. \quad (\text{h})$$

The summation is equivalent to $(\mathbf{B}^\top)^{-1}$. Performing this substitution results in

$$\text{Var}[\mathbf{Y}_n] = \mathbf{B} (\mathbf{B}^\top)^{-1} \mathbf{B}^\top \quad (\text{i})$$

$$= \mathbf{B} \quad (\text{j})$$

By eq. (a), $\mathbf{B}$ is given by

$$\left(\sum_{i=1}^r \mathbf{A}_i \right)^{-1} \quad (\text{k})$$

and by eq. (5.22), $\mathbf{A}_i$ is given by

$$n_i \mathcal{I}(\boldsymbol{\theta}^* | w_i). \quad (\text{l})$$

Performing these substitution results in

$$\text{Var}[\mathbf{Y}_n] = \left(\sum_{i=1}^r n_i \mathcal{I}(\boldsymbol{\theta}^* | w_i) \right)^{-1}. \quad (\text{m})$$

□

The weighted estimator asymptotically achieves the Cramér-Rao lower-bound as given by eq. (5.5) thus it is the asymptotic UMVUE estimator of θ^* given a random sample of masked system failure times in which n_i realizations have w_i candidates for $i = 1, \dots, r$.

A linear combination of multivariate normal distributions is a multivariate normal distribution, thus the asymptotic sampling distribution of $\hat{\theta}_n$ is normally distributed.

Postulate 5.5. *As $n \rightarrow \infty$, the sampling distribution of $\hat{\theta}_n$ converges in distribution to a multivariate normal with a mean θ^* and a variance-covariance given by eq. (5.23), written*

$$\mathbf{Y}_n \xrightarrow{d} \text{MVN} \left(\theta^*, \left(\sum_{i=1}^r n_i \mathcal{I}(\theta^* | w_i) \right)^{-1} \right). \quad (5.24)$$

The generative model for $\mathbf{Y}_n$ is given by algorithm 4.

Algorithm 3: Generative model of maximum likelihood estimator

Input:

Θ^* , the true parameter value of the series system.

Output:

$\hat{\theta}_n$, a realization of $\mathbf{Y}_n$.

```

1 Model generate_mle( $\Theta^*$ )
2   draw candidate cardinality  $w \sim p_W(\cdot)$ 
3    $\hat{\theta}_n \leftarrow \text{generate\_mle}(\Theta^*, w)$ 
4   return  $\hat{\theta}_n$ 

```

5.2.1 Bootstrap of sample covariance

Another estimator of the variance-covariance of $\mathbf{Y}_n$ is given by the sample covariance.

Definition 5.6. *Given a sample of r maximum likelihood estimates, $\hat{\theta}_n^{(1)}, \dots, \hat{\theta}_n^{(r)}$, the maximum likelihood estimator of the variance-covariance of the sampling distribution of $\hat{\theta}_n$ is given by the sample covariance,*

$$\hat{\text{Var}}[\mathbf{Y}_n] = \frac{1}{r} \sum_{i=1}^r \left(\hat{\theta}_n^{(i)} - \theta^* \right) \cdot \left(\hat{\theta}_n^{(i)} - \theta^* \right)^\top. \quad (5.25)$$

The sample covariance depends upon a random sample of maximum likelihood estimates, and thus the sample covariance is a random matrix. However, by the property that maximum likelihood estimators converge in probability to the true value,

$$\hat{\text{Var}}[\mathbf{Y}_n] \xrightarrow{P} \text{Var}[\mathbf{Y}_n], \quad (5.26)$$

it follows that

$$\mathbf{Y}_n \xrightarrow{d} \text{MVN} \left(\theta^*, \hat{\text{Var}}[\mathbf{Y}_n] \right). \quad (5.27)$$

If only one maximum likelihood estimate is realized, $\mathbf{Y}_n = \hat{\theta}_n$, then the sample covariance given by ?? cannot be computed. In the *parametric Bootstrap*, the sample covariance is approximated by using $\hat{\theta}_n$ as an estimate of θ^* in the generative model described by ??, i.e.,

$$\hat{\theta}_n^{(1)}, \dots, \hat{\theta}_n^{(r)} \leftarrow \text{generate_mle}(\hat{\theta}_n), \quad (5.28)$$

and the sample covariance of these approximate maximum likelihood estimates is an asymptotically unbiased estimator of $\text{Var} [\mathbf{Y}_n]$.

Algorithm 4: Sample covariance of a bootstrapped sample of maximum likelihood estimates.

Input:

$\hat{\boldsymbol{\theta}}_n$, an estimate of $\boldsymbol{\theta}^*$.

Output:

sample covariance of a bootstrapped sample of maximum likelihood estimates.

1 **Model** *BootstrapCovariance*

2 | $(\hat{\boldsymbol{\theta}}_n)$

3 **for** $i = 1$ to r **do**

4 | $\hat{\boldsymbol{\theta}}_n^{(i)} \leftarrow \text{generate_mle}(\hat{\boldsymbol{\theta}}_n)$

5 **end**

6 **return** sample covariance of $\hat{\boldsymbol{\theta}}_n^{(1)}, \dots, \hat{\boldsymbol{\theta}}_n^{(r)}$

6 Sampling distribution of functions of parameters

Suppose we have a characteristic of interest $\mathbf{g}: \mathbb{R}^{m \cdot q} \mapsto \mathbb{R}^p$ that is a function of the parametric model. The *true* value of the characteristic is given by $\mathbf{g}(\boldsymbol{\theta}^*)$. By the invariance property of maximum likelihood estimators, if $\hat{\boldsymbol{\theta}}_n$ is the maximum likelihood estimator of $\boldsymbol{\theta}^*$ then $\mathbf{g}(\hat{\boldsymbol{\theta}}_n)$ is the maximum likelihood estimator of $\mathbf{g}(\boldsymbol{\theta}^*)$.

By postulate 5.4, $\mathbf{Y}_n$ is a random vector drawn from a multivariate normal distribution,

$$\mathbf{Y}_n \sim \text{MVN}(\boldsymbol{\theta}^*, \text{Var}[\mathbf{Y}_n]) , \quad (5.24 \text{ revisited})$$

therefore $\mathbf{g}(\mathbf{Y}_n)$ is a random vector. Under the regularity conditions (see assumption 8), $\mathbf{g}(\mathbf{Y}_n)$ is asymptotically normally distributed with a mean given by $\mathbf{g}(\boldsymbol{\theta}^*)$ and a variance-covariance given by $\text{Var}[\mathbf{g}(\mathbf{Y}_n)]$, written

$$\mathbf{g}(\mathbf{Y}_n) \xrightarrow{d} \text{MVN}(\mathbf{g}(\boldsymbol{\theta}^*), \text{Var}[\mathbf{g}(\mathbf{Y}_n)]) . \quad (6.1)$$

6.1 Confidence intervals

A property of multivariate normal distributions is that the marginal of any subset of the components is given by dropping the irrelevant components from the mean vector $\boldsymbol{\theta}^*$ and the variance-covariance $\text{Var}[\mathbf{Y}_n]$.

If we are interested in the asymptotic marginal distribution of the j^{th} component, we drop all of the other components, resulting in the univariate normal distribution given by

$$[\mathbf{Y}_n]_j \sim \mathcal{N}\left(\theta_j^*, \frac{\sigma^2}{n}\right) , \quad (6.2)$$

where $\sigma^2 = [\mathcal{I}^{-1}(\boldsymbol{\theta}^*)]_{jj}$.

Asymptotically, the random variable $[\mathbf{Y}_n]_j$ realizes a value less than any constant a with a probability given by

$$P\left[[\mathbf{Y}_n]_j < a\right] = \Phi\left(\frac{a - \Theta_j^*}{\sigma_j}\right), \quad (6.3)$$

where Φ is the cumulative distribution function of the standard normal.

The smallest interval in which $[\mathbf{Y}_n]_j$ is realized with probability p is given by

$$\theta_j^* \pm \sigma_j \Phi^{-1}(p/2), \quad (6.4)$$

where Φ^{-1} is the inverse cumulative distribution function of the standard normal.

By the invariance property of maximum likelihood estimators, a maximum likelihood estimator of eq. (6.4) is the confidence interval given by the following definition.

Definition 6.1. *An asymptotic $(1 - \alpha) \cdot 100\%$ confidence interval for the j^{th} component of the true parameter index θ^* is given by*

$$\hat{\theta}_j \pm \hat{\sigma}_j \Phi^{-1}(1 - \alpha/2), \quad (6.5)$$

where $\hat{\theta}_j$ is the j^{th} component of $\hat{\boldsymbol{\theta}}_n$ and $\hat{\sigma}_j$ is the j^{th} diagonal element of $\hat{\text{Var}}[\mathbf{Y}_n]$.

Note that the confidence interval given by eq. (6.5) is a realization of a random interval since $\hat{\theta}_j$ is a realization of the normal distribution given by eq. (6.2) and $\hat{\sigma}_j$ is a function $\hat{\theta}_j$, where it is expected that $(1 - \alpha/2) \cdot 100\%$ of the random intervals generated contain the true parameter index θ_j^* .

6.2 Hypothesis testing

The intervals discussed in section 6.1 ignore correlations between the components of $\mathbf{Y}_n$. Asymptotically, $\mathbf{Y}_n$ has a probability p of occurring within the hyper-ellipsoid (the smallest hyper-volume with probability p) centered around $\boldsymbol{\theta}^*$ given by

$$(\mathbf{Y}_n - \boldsymbol{\theta}^*)^\top \mathcal{I}_n(\boldsymbol{\theta}^*) (\mathbf{Y}_n - \boldsymbol{\theta}^*) \leq \chi_{m \cdot q}^2(p), \quad (6.6)$$

where $\chi_{m \cdot q}^2(p)$ is the p quantile of the chi-square distribution with $m \cdot q$ degrees of freedom.

By the invariance property of maximum likelihood estimators, a maximum likelihood estimator of eq. (6.6) is given by the confidence region.

Definition 6.2. *Suppose we have a maximum likelihood estimate $\hat{\boldsymbol{\theta}}_n$ and hypothesize that the true parameter index is given by $\boldsymbol{\theta}$. At significance level α we fail to reject the null hypothesis $H_0: \boldsymbol{\theta}^* = \boldsymbol{\theta}$ versus the alternative hypothesis $H_a: \boldsymbol{\theta}^* \neq \boldsymbol{\theta}$ if*

$$(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_n)^\top \mathcal{I}_n(\hat{\boldsymbol{\theta}}_n) (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_n) \leq \chi_{m \cdot q}^2(1 - \alpha), \quad (6.7)$$

where $\chi_{m \cdot q}^2(1 - \alpha)$ is the $(1 - \alpha)$ quantile of the chi-square distribution with $m \cdot q$ degrees of freedom.

Applying eq. (6.7) to the marginal of $[\mathbf{Y}(n|w)]_j$ generates the confidence interval described in ??.

6.3 Monte-carlo simulation

The generative model of the sampling distribution of $\mathbf{Y}_n$, as described in algorithm 4, is asymptotically equivalent to generating samples from the multivariate normal distribution given by ???. However, generating samples from the multivariate normal is much less computationally demanding than the non-asymptotic model.

The generative model may be used to generate samples of any statistic that is a function of the true parameter index θ^* . A sample drawn from $\mathbf{g}(\mathbf{Y}(n|w))$ may be generated by drawing r maximum likelihood estimates from the sampling distribution

$$\hat{\theta}_n^{(1)}, \dots, \hat{\theta}_n^{(r)}, \quad (6.8)$$

and applying $\mathbf{g}$ to each, resulting in the sample

$$\mathbf{g}(\hat{\theta}_n^{(1)}), \dots, \mathbf{g}(\hat{\theta}_n^{(r)}). \quad (6.9)$$

An estimate of the variance-covariance $\text{Var}[\mathbf{g}(\mathbf{Y}_n)]$ is given by the sample covariance

$$\begin{aligned} \hat{\text{Var}}[\mathbf{g}(\mathbf{Y}_n)] = \\ \frac{1}{r} \sum_{i=1}^r \left(\mathbf{g}(\hat{\theta}_n^{(i)}) - \mathbf{g}(\hat{\theta}_n) \right) \left(\mathbf{g}(\hat{\theta}_n^{(i)}) - \mathbf{g}(\hat{\theta}_n) \right)^\top. \end{aligned} \quad (6.10)$$

Thus,

$$\mathbf{g}(\mathbf{Y}_n) \sim \text{MVN} \left(\mathbf{g}(\hat{\theta}_n), \hat{\text{Var}}[\mathbf{Y}_n] \right). \quad (6.11)$$

In what follows, we explore a couple of characteristics of interest about the series system.

6.3.1 Mean time to failure of components

The *mean time to failure* of component j is given by

$$\mathbf{g}(\theta^*) = \mathbb{E}_{\theta^*}[\mathbf{T}_j]. \quad (6.12)$$

Suppose $\mathbf{g}(\theta^*) \in \mathbb{R}^{m \cdot q}$, then the variance-covariance of

$$\mathcal{J}(\mathbf{g}(\theta^*)) \mathcal{I}_n^{-1}(\theta^*) \mathcal{J}(\mathbf{g}(\theta^*))^\top, \quad (6.13)$$

where $\mathcal{J}(\cdot) \in \mathbb{R}^?$ is the Jacobian as given by ??.

The maximum likelihood estimator of θ^* is given by $\mathbf{g}(\hat{\theta}_n)$ and is approximately normally distributed,

$$\mathbf{g}(\mathbf{Y}_n) \xrightarrow{d} \mathcal{N} \left(\mathbf{g}(\hat{\theta}_n), \hat{\sigma}^2 \right). \quad (6.14)$$

where $\hat{\sigma}^2$ is the sample variance.

7 The maximum entropy distribution: exponentially distributed component lifetimes

Suppose we have a series systems in which component lifetimes are exponentially distributed.

7.1 Parametric functions

Suppose the j^{th} component has a failure rate λ_j^* , denoted by $T_j \sim \text{EXP}(\lambda_j^*)$. The random variable T_j has reliability, density, and failure rate functions given respectively by

$$R_j(t | \lambda_j^*) = \exp(-\lambda_j^* \cdot t), \quad (7.1)$$

$$f_j(t | \lambda_j^*) = \lambda_j^* \exp(-\lambda_j^* \cdot t), \quad (7.2)$$

$$h_j(\cdot | \lambda_j^*) = \lambda_j^*, \quad (7.3)$$

where $t > 0$ and $\lambda_j^* > 0$.

A series system with exponentially distributed lifetimes is also exponentially distributed.

Theorem 7.1. *The random lifetime S of a series system composed of m components with exponentially distributed lifetimes is exponentially distributed with a failure rate that is the sum of the component failure rates with a true parameter index given by*

$$\boldsymbol{\lambda}^* = \langle \lambda_1^*, \dots, \lambda_m^* \rangle^T \quad (7.4)$$

where λ_j^* is the failure rate of the j^{th} component. That is,

$$S \sim \text{EXP} \left(\sum_{j=1}^m \lambda_j^* \right), \quad (7.5)$$

Proof. By theorem 3.1,

$$R_S(t | \boldsymbol{\lambda}^*) = \prod_{j=1}^m R_j(t | \lambda_j^*). \quad (a)$$

Plugging in the component reliability functions as given by eq. (7.1) results in

$$R_S(t | \boldsymbol{\lambda}^*) = \prod_{j=1}^m \exp(-\lambda_j^* \cdot t) = \exp \left(- \left[\sum_{j=1}^m \lambda_j^* \right] t \right). \quad (b)$$

This reliability function belongs to the family of exponential distributions with a true parameter index given by the sum of the component failure rates. $\square$

By theorem 7.1, the system has a probability density, failure rate, and reliability functions given respectively by

$$f_S(t | \boldsymbol{\lambda}^*) = \left(\sum_{j=1}^m \lambda_j^* \right) \exp \left(- \left[\sum_{j=1}^m \lambda_j^* \right] t \right), \quad (7.6)$$

$$h_S(\cdot | \boldsymbol{\lambda}^*) = \sum_{j=1}^m \lambda_j^*, \quad (7.7)$$

$$R_S(t | \boldsymbol{\lambda}^*) = \exp \left(- \left[\sum_{j=1}^m \lambda_j^* \right] t \right) \quad (7.8)$$

where $t > 0$ and $\lambda_j^* > 0$.

The conditional probability that component k is the cause of a system failure at time t is given by

$$p_{K|S}(k|t, \boldsymbol{\lambda}^*) = p_K(k|\boldsymbol{\lambda}^*) = \frac{\lambda_k^*}{\sum_{p=1}^m \lambda_p^*} \mathbb{1}_{\mathcal{K}}(k). \quad (7.9)$$

Proof. By theorem 3.3,

$$p_{K|S}(k|t, \boldsymbol{\lambda}^*) = \frac{h_k(t|\boldsymbol{\lambda}_k^*)}{h_S(t|\boldsymbol{\lambda}^*)}. \quad (a)$$

Plug in the failure rate of component k and the failure rate of the system given by eqs. (7.3) and (7.7). $\square$

Due to the constant failure rates of the components, K and S are independent. The joint density of K and S is given by

$$f_{K,S}(k, t|\boldsymbol{\lambda}^*) = \lambda_k^* \exp\left(-\left[\sum_{j=1}^m \lambda_j^*\right] t\right) \mathbb{1}_{\mathcal{K} \times \mathbb{R}^+}(k, t) \quad (7.10)$$

Proof. By definition,

$$f_{K,S}(k, t|\boldsymbol{\lambda}^*) = p_{K|S}(k|t, \boldsymbol{\lambda}^*) f_S(t|\boldsymbol{\lambda}^*). \quad (a)$$

Plug in the conditional probability and the marginal probability given by eqs. (7.6) and (7.9). $\square$

The conditional joint density of C and S given $W = w$ is given by

$$f_{C,S|W}(\mathcal{C}, t|w, \boldsymbol{\lambda}^*) = \frac{\mathbb{1}_{t>0} \mathbb{1}_{|\mathcal{C}|=w}}{\binom{m-1}{w-1}} \left(\sum_{j \in \mathcal{C}} \lambda_j\right) \exp\left[-\left(\sum_{j=1}^m \lambda_j\right) t\right], \quad (7.11)$$

whose proof follows from corollary 3.7.1.

The conditional probability that a component in C is the cause of a system failure given that $S = t$ and $W = w$ is given by

$$p_{C|S,W}(\mathcal{C}|t, w, \boldsymbol{\lambda}^*) = \frac{\sum_{j \in \mathcal{C}} \lambda_j^*}{\binom{m-1}{w-1} \sum_{p=1}^m \lambda_p^*} \mathbb{1}_{\mathbb{R}_+ \times \mathcal{W}_w}(t, |\mathcal{C}|). \quad (7.12)$$

whose proof follows from corollary 3.7.2.

Note that the marginal distribution of $C|W = w$ is given by

$$p_{C|W}(\mathcal{C}|w, \boldsymbol{\lambda}^*) = \frac{\sum_{j \in \mathcal{C}} \lambda_j^*}{\binom{m-1}{w-1} \sum_{p=1}^m \lambda_p^*} \mathbb{1}_{\mathcal{W}_w}(|\mathcal{C}|), \quad (7.13)$$

Proof. The marginal is given by

$$p_{C|W}(\mathcal{C}|w, \boldsymbol{\lambda}^*) = \int_{-\infty}^{\infty} f_{C,S|W}(\mathcal{C}, t|w, \boldsymbol{\lambda}^*) dt \quad (a)$$

$$= \int_0^{\infty} \frac{\mathbb{1}_{\mathbb{R}_+ \times \mathcal{W}_w}(t, |\mathcal{C}|)}{\binom{m-1}{w-1}} \left(\sum_{j \in \mathcal{C}} \lambda_j\right) \exp\left[-\left(\sum_{j=1}^m \lambda_j\right) t\right] dt \quad (b)$$

$$= \frac{\mathbb{1}_{\mathcal{W}_w}(|\mathcal{C}|)}{\binom{m-1}{w-1}} \sum_{j \in \mathcal{C}} \lambda_j \int_0^{\infty} \exp\left[-\left(\sum_{j=1}^m \lambda_j\right) t\right] dt. \quad (c)$$

Multiplying the integrand by

$$\frac{\sum_{j=1}^m \lambda_j}{\sum_{j=1}^m \lambda_j} \quad (d)$$

results in

$$p_{C|W}(C|w, \lambda^*) = \frac{\mathbb{1}_{\mathcal{W}_w(\{C\})}}{\binom{m-1}{w-1}} \frac{\sum_{j \in C} \lambda_j}{\sum_{j=1}^m \lambda_j} \int_0^\infty \left(\sum_{j=1}^m \lambda_j \right) \exp \left[- \left(\sum_{j=1}^m \lambda_j \right) t \right] dt \quad (e)$$

$$= \frac{\mathbb{1}_{\mathcal{W}_w(\{C\})}}{\binom{m-1}{w-1}} \frac{\sum_{j \in C} \lambda_j}{\sum_{j=1}^m \lambda_j} \int_0^\infty f_S(t | \lambda^*) dt \quad (f)$$

$$= \frac{\mathbb{1}_{\mathcal{W}_w(\{C\})}}{\binom{m-1}{w-1}} \frac{\sum_{j \in C} \lambda_j}{\sum_{j=1}^m \lambda_j} . \quad (g)$$

□

The marginal distribution of $C|W = w$ given by ?? is the same as the joint distribution of $C, S|W = w$ given by eq. (7.13). Thus, C and S are conditionally independent given $W = w$.

7.2 Maximum entropy distribution

The larger the variance of the estimator, the larger a sample necessary to obtain an estimate with a given mean squared error.

Entropy is a fundamental measure of uncertainty. If a diagnostician could ask “yes” or “no” questions about the component cause of failure, an *expected lower-bound* on the number of questions required to determine which component caused the failure is given by

$$H(K) = - \sum_{k \in \mathcal{K}} p_K(k) \log_2 p_K(k) . \quad (7.14)$$

This is known as the *entropy* of the discrete random variable K , which quantifies the amount of uncertainty about which value K will realize.

The greater the entropy, the greater the number of questions the diagnostician needs on average to determine the component cause of failure. Each component may have a different distribution of lifetimes, and thus knowing when a system failure occurs may *reduce* the uncertainty about the component cause of failure,

$$H(K|S) \leq H(K) . \quad (7.15)$$

If K and S are *statistically independent*, then

$$H(K|S) = H(K) , \quad (7.16)$$

that is, observing the system failure time does not reduce the uncertainty. The joint entropy of K and S is given by

$$H(K, S) = - \sum_{k \in \mathcal{K}} f_{K,S}(k, t) \log_2 f_{K,S}(k, t) , \quad (7.17)$$

which is equivalent to

$$H(K|S) + H(S) . \quad (7.18)$$

Given that $H(K|S) \leq H(K)$, a maximum entropy distribution has the form

$$H(K, S) = H(K) + H(S) , \quad (7.19)$$

where each of the components, and thus the system lifetime S , has an interval of support $(0, \infty)$.

Given the way in which K is defined as a function of S , there is a possibility that S and K are necessarily dependent. However, by ??, we see that if the component lifetimes are exponentially distributed, then the system lifetime S is exponentially distributed, and by ?? K is *independent* of S .

Moreover, if we constrain the system lifetime to have an expectation μ , then the maximum entropy distribution of S is given by

$$S \sim \text{EXP}(\lambda = 1/\mu). \quad (7.20)$$

Independently, the maximum entropy distribution of the discrete random variable K is given by the discrete uniform distribution

$$K \sim \text{UNIF}(1, m). \quad (7.21)$$

By ??, we know that these components must be exponentially distributed and that each component must be uniformly distributed such the sum of their failure rates must equal λ , i.e.,

$$T_j \sim \text{EXP}(\lambda/m) \quad (7.22)$$

for $j = 1, \dots, m$.

TODO: fix this

The entropy of $\langle S, K \rangle$ has very little to do determining the *parameters*. The preceding discussion only applied to the number of questions, on average, needed to discern which component failed, and, independently, when did it fail, i.e., sample from K and then from S

7.3 Maximum likelihood estimator

According to definition 5.3, the maximum likelihood estimator of λ^* is the value that maximizes the log-likelihood on a given masked system failure time sample, denoted $\mathbf{M}_n$. The likelihood function is given by

$$\mathcal{L}(\lambda | \mathbf{M}_n) \propto \exp \left[- \left(\sum_{j=1}^m \lambda_j \right) \left(\sum_{i=1}^n t_i \right) \right] \left[\prod_{i=1}^n \left(\sum_{j \in \mathcal{C}_i} \lambda_j \right) \right], \quad (7.23)$$

whose proof follows from definition 4.3. Therefore, the log-likelihood function ℓ is given by

$$\ell(\lambda | \mathbf{M}_n) = \text{const} + \sum_{i=1}^n \ln \left[\sum_{j \in \mathcal{C}_i} \lambda_j \right] - \left[\sum_{i=1}^n t_i \right] \left[\sum_{j=1}^m \lambda_j \right]. \quad (7.24)$$

Thus, maximum likelihood estimator of λ^* is given by

$$\hat{\lambda}_n = \arg \max_{\lambda > 0} \ell(\lambda | \mathbf{M}_n). \quad (7.25)$$

The solution to eq. (7.25) must be a stationary point, i.e., a point at which the score function is zero. The j^{th} component of the score function is given by

$$\frac{\partial \ell}{\partial \lambda_j} = \sum_{i=1}^n \left(\sum_{p \in \mathcal{C}_i} \lambda_p \right)^{-1} \mathbb{1}_{\mathcal{C}_i}(j) - \sum_{i=1}^n t_i. \quad (7.26)$$

According to postulate 5.4, the sampling distribution of $\hat{\lambda}_{\mathbf{n}}$ converges in distribution to a multivariate normal with a mean given by λ^* and a variance-covariance given by

$$\frac{1}{n} \mathcal{I}^{-1}(\lambda^*). \quad (7.27)$$

Since λ^* is not known it may be approximated by

$$\mathcal{I}^{-1}(\hat{\lambda}_{\mathbf{n}}). \quad (7.28)$$

The (j, k) -th element of the Fisher information matrix is given by

$$[\mathcal{I}(\lambda^*)]_{jk} = \frac{\sum_{\mathcal{C}} \left[\left(\sum_{p \in \mathcal{C}} \lambda_p^* \right)^{-1} \mathbb{1}_{\mathcal{C} \times \mathcal{C}}(j, k) \right]}{\binom{m-1}{w-1} \sum_{p=1}^m \lambda_p^*}. \quad (7.29)$$

where the summation indexed by $\mathcal{C}$ is over $2^{\mathcal{K}}$.

Proof. By eq. (4.11), the (j, k) -th element of the information matrix is given by

$$[\mathcal{I}(\lambda^*)]_{ij} = - \sum_{\mathcal{C}} \int_0^\infty \frac{\partial^2}{\partial \lambda_i \partial \lambda_j} \ln f_{\mathcal{C}, S | W}(\mathcal{C}, t | w, \text{mat}(\lambda)) \bigg|_{\lambda=\lambda^*} \cdot f_{\mathcal{C}, S | W}(\mathcal{C}, t | w, \text{mat}(\lambda^*)) dt, \quad (a)$$

In the above equation, the second-order partial derivative of the log of the joint density is equivalent to the partial derivative with respect to the k^{th} parameter of the j^{th} component of the score, given by eq. (7.26), conditioned on a masked system failure time sample of size 1, denoted by $\mathbf{M}_{\mathbf{n}1}$. This mixed second-order partial derivative is given by

$$\frac{\partial}{\partial \lambda_k} \frac{\partial \ell}{\partial \lambda_j} = \frac{\partial}{\partial \lambda_k} \left(\left[\sum_{p \in \mathcal{C}} \lambda_p \right]^{-1} \mathbb{1}_{\mathcal{C}}(j) - t \right) \quad (b)$$

$$= - \left(\sum_{p \in \mathcal{C}} \lambda_p \right)^{-2} \mathbb{1}_{\mathcal{C} \times \mathcal{C}}(j, k). \quad (c)$$

Plugging in the joint density and substituting eq. (c) into eq. (a) results in

$$[\mathcal{I}(\lambda^*)]_{ij} = \sum_{\mathcal{C}} \int_0^\infty \left(\sum_{p \in \mathcal{C}} \lambda_p \right)^{-2} \mathbb{1}_{\mathcal{C} \times \mathcal{C}}(p, k) \bigg|_{\lambda=\lambda^*} \frac{1}{\binom{m-1}{w-1}} \left(\sum_{p \in \mathcal{C}} \lambda_p^* \right) \exp \left(- \left[\sum_{p=1}^m \lambda_p^* \right] t \right) dt. \quad (d)$$

Simplifying and rearranging results in

$$[\mathcal{I}(\lambda^*)]_{ij} = \frac{1}{\binom{m-1}{w-1}} \sum_{\mathcal{C}} \left(\sum_{p \in \mathcal{C}} \lambda_p^* \right)^{-1} \mathbb{1}_{\mathcal{C} \times \mathcal{C}}(p, k) \int_0^\infty \exp \left(- \left[\sum_{p=1}^m \lambda_p^* \right] t \right) dt. \quad (e)$$

Multiplying the integral by $\frac{\sum_{p=1}^m \lambda_p^*}{\sum_{p=1}^m \lambda_p^*}$ results in an integral of the form

$$\int_0^\infty \frac{\sum_{p=1}^m \lambda_p^*}{\sum_{p=1}^m \lambda_p^*} \exp \left(- \left[\sum_{p=1}^m \lambda_p^* \right] t \right) dt. \quad (f)$$

Pulling the $\left(\sum_{p=1}^m \lambda_p^*\right)^{-1}$ constant out of the integrand results in

$$\frac{1}{\sum_{p=1}^m \lambda_p^*} \int_0^\infty \left(\sum_{p=1}^m \lambda_p^*\right) \exp\left(-\left[\sum_{p=1}^m \lambda_p^*\right] t\right) dt = \frac{1}{\sum_{p=1}^m \lambda_p^*} \int_0^\infty f_S(t | \boldsymbol{\lambda}^*) dt \quad (\text{g})$$

$$= \frac{1}{\sum_{p=1}^m \lambda_p^*}. \quad (\text{h})$$

Substituting eq. (h) into eq. (e) results in

$$[\mathcal{I}(\boldsymbol{\lambda}^*)]_{ij} = \frac{1}{\binom{m-1}{w-1}} \sum_{\mathcal{C}} \left\{ \left(\sum_{p \in \mathcal{C}} \lambda_p^*\right)^{-1} \frac{1}{\sum_{p=1}^m \lambda_p^*} \mathbb{1}_{\mathcal{C} \times \mathcal{C}(p, k)} \right\}. \quad (\text{i})$$

□

7.3.1 Sufficient statistics

Consider a sample of n masked system failure times denoted by $\mathbf{m}_n$. The mean system lifetime of the sample is given by

$$\bar{t} = \frac{1}{n} \sum_{i=1}^n t_i. \quad (7.30)$$

and another statistic, denoted by $\hat{\omega}$, is a dictionary of candidate set-frequency count pairs where a candidate set $\mathbf{k} \in 2^{\mathcal{K}}$ maps to its sample frequency and is given by

$$\hat{\omega}_{\mathbf{k}} = \sum_{i=1}^n \mathbb{1}_{\mathbf{k}}(\mathcal{C}_i). \quad (7.31)$$

Given a sample $\mathbf{m}_n$, the likelihood $\mathcal{L}(\boldsymbol{\lambda} | \mathbf{m}_n)$ described by eq. (7.23) is the same as the likelihood given by

$$\mathcal{L}(\boldsymbol{\lambda} | \bar{t}, \hat{\omega}) = \exp\left(-n\bar{t} \sum_{j=1}^m \lambda_j\right) \prod_{\mathcal{C} \in \mathcal{W}_w} \left(\sum_{j \in \mathcal{C}} \lambda_j\right)^{\hat{\omega}_{\mathcal{C}}}, \quad (7.32)$$

where $\mathcal{W}_w$ is the set of all candidate sets of cardinality w .

Proof. The likelihood function $\mathcal{L}$ with respect to a sample of n masked system failure times is given by

$$\mathcal{L}(\boldsymbol{\lambda} | \mathbf{m}_n) \propto \exp\left[-\left(\sum_{j=1}^m \lambda_j\right) \left(\sum_{i=1}^n t_i\right)\right] \left[\prod_{i=1}^n \left(\sum_{j \in \mathcal{C}_i} \lambda_j\right)\right]. \quad (7.23 \text{ revisited})$$

Substituting $\sum_{i=1}^n t_i$ in the above equation with the equivalent expression $n\bar{t}$ results in

$$\mathcal{L}(\boldsymbol{\lambda} | \mathbf{M}_n) \propto \exp\left[-n\bar{t} \sum_{j=1}^m \lambda_j\right] \underbrace{\left[\prod_{i=1}^n \left(\sum_{j \in \mathcal{C}_i} \lambda_j\right)\right]}_A. \quad (\text{a})$$

In the above equation, each unique set $(\lambda_{j_1} + \dots + \lambda_{j_w})$ in the product (labeled A) has multiplicity $\hat{\omega}_{\{\lambda_{j_1} \dots \lambda_{j_w}\}}$, thus we may substitute this product with the equivalent product

$$\prod_{\mathcal{C} \in \mathcal{W}_w} \left(\sum_{j \in \mathcal{C}} \lambda_j \right)^{\hat{\omega}_{\mathcal{C}}}, \quad (\text{b})$$

where $\mathcal{C}$ is over all unique candidate sets of cardinality w . The result of this substitution is given by

$$\mathcal{L}(\boldsymbol{\lambda} | \bar{t}, \hat{\omega}) = \exp \left(-n\bar{t} \sum_{j=1}^m \lambda_j \right) \prod_{\mathcal{C} \in \mathcal{W}_w} \left(\sum_{j \in \mathcal{C}} \lambda_j \right)^{\hat{\omega}_{\mathcal{C}}}. \quad (\text{c})$$

□

The statistics $\bar{t}$ and $\hat{\omega}$ are jointly sufficient for $\boldsymbol{\lambda}^*$

Proof. By the Fisher–Neyman factorization theorem, since $\mathcal{L}$ given by eq. (7.23) can be factored as $\mathcal{L}(\bar{t}, \hat{\omega}) = h(\bar{t}, \hat{\omega})g(\bar{t}, \hat{\omega})$, $\hat{\omega}$ and $\bar{t}$ are joint sufficient statistics for $\boldsymbol{\lambda}^*$. □

It may be helpful to consider the following example.

Example 1 Suppose we have a 3-out-of-3 system with exponentially distributed component lifetimes. If we are given a sample of 4 masked system times,

$$\mathbf{m}_4 = (\langle t_1 = 1, \mathcal{C}_1 = \{1, 2\} \rangle, \langle t_2 = 1, \mathcal{C}_2 = \{1, 3\} \rangle, \langle t_3 = 2, \mathcal{C}_3 = \{1, 2\} \rangle, \langle t_4 = 2, \mathcal{C}_4 = \{1, 2\} \rangle), \quad (\text{a})$$

then joint sufficient statics of $\boldsymbol{\lambda}^*$ are

$$\begin{aligned} \bar{t} &= \frac{1 + 1 + 2 + 2}{4} = \frac{3}{2}, \\ \hat{\omega} &= \{\hat{\omega}_{\{1,2\}} \mapsto 3, \hat{\omega}_{\{1,3\}} \mapsto 1\}. \end{aligned}$$

No additional information is needed to compute the likelihood of $\boldsymbol{\lambda}$ with respect to the given $\mathbf{M}_4$ sample.

The log-likelihood ℓ is given by

$$\ell(\boldsymbol{\theta} | \bar{t}, \hat{\omega}, w) = -n\bar{t} \left(\sum_{j=1}^m \lambda_j \right) + \sum_{\mathcal{C} \in \mathcal{W}_w} \hat{\omega}_{\mathcal{C}} \ln \left(\sum_{j \in \mathcal{C}} \lambda_j \right). \quad (7.33)$$

To find the MLE $\hat{\boldsymbol{\lambda}}_n$, we find the stationary points for $\lambda_1, \dots, \lambda_m$ given by

$$0 = -n\bar{t} + \sum_{\mathcal{C} \in \mathcal{W}_w} \hat{\omega}_{\mathcal{C}} \left(\sum_{j \in \mathcal{C}} \lambda_j \right)^{-1} \mathbb{1}_{\mathcal{C}}(k); k = 1, \dots, m. \quad (7.34)$$

The (j, k) -th element of the observed information matrix $\mathcal{J}_n$ is given by

$$[\mathcal{J}_n(\boldsymbol{\lambda}^* \hat{\omega})]_{jk} = - \sum_{\mathcal{C} \in \mathcal{W}_w} \hat{\omega}_{\mathcal{C}} \left(\sum_{i \in \mathcal{C}} \lambda_i^* \right)^{-2} \mathbb{1}_{\mathcal{C} \times \mathcal{C}}(j, k), \quad (7.35)$$

which only depends on the statistic $\hat{\omega}$.

7.4 Applications

A $(1 - \alpha) \cdot 100\%$ -confidence interval for λ_j^* is given by

$$\hat{\lambda}_j \pm z_{1-\alpha/2} \sqrt{\frac{1}{n} [\mathcal{I}^{-1}(\hat{\lambda}_{\mathbf{n}})]_{jj}}. \quad (7.36)$$

7.5 Case study: 3-out-of-3 system

The 3-out-of-3 system's lifetime distribution functions are given by

$$R_S(t | \boldsymbol{\lambda}^*) = \exp [-(\lambda_1^* + \lambda_2^* + \lambda_3^*)t], \quad (7.37)$$

$$f_S(t | \boldsymbol{\lambda}^*) = (\lambda_1^* + \lambda_2^* + \lambda_3^*) \exp [-(\lambda_1^* + \lambda_2^* + \lambda_3^*)t], \quad (7.38)$$

$$h_S(t | \boldsymbol{\lambda}^*) = \lambda_1^* + \lambda_2^* + \lambda_3^*, \quad (7.39)$$

where $t > 0$, $\lambda_1^* > 0$, $\lambda_2^* > 0$, and $\lambda_3^* > 0$.

7.5.1 Sampling distributions of the maximum likelihood estimator

Component sets of size two

If we condition on samples in which each observation consists of two candidates, $W = |C| = 2$, the likelihood function is given by

$$\mathcal{L}(\boldsymbol{\lambda} | \bar{t}, \hat{\omega}) = (\lambda_1 + \lambda_2)^{\hat{\omega}_{\{1,2\}}} (\lambda_1 + \lambda_3)^{\hat{\omega}_{\{1,3\}}} (\lambda_2 + \lambda_3)^{\hat{\omega}_{\{2,3\}}} \exp \left(-n\bar{t}(\lambda_1 + \lambda_2 + \lambda_3) \right) \quad (7.40)$$

and the log-likelihood function is given by

$$\begin{aligned} \ell(\boldsymbol{\lambda} | \bar{t}, \hat{\omega}) = & \hat{\omega}_{\{1,2\}} \ln(\lambda_1 + \lambda_2) + \hat{\omega}_{\{1,3\}} \ln(\lambda_1 + \lambda_3) + \\ & \hat{\omega}_{\{2,3\}} \ln(\lambda_2 + \lambda_3) - n\bar{t}(\lambda_1 + \lambda_2 + \lambda_3). \end{aligned} \quad (7.41)$$

The maximum likelihood estimate $\hat{\lambda}_{\mathbf{n}}$ is given by the solving the system of equations in which the score is zero, $s(\boldsymbol{\lambda}) = \mathbf{0}$, where the score is given by

$$s(\boldsymbol{\lambda}) = \begin{pmatrix} \frac{\hat{\omega}_{\{1,2\}}}{\lambda_1 + \lambda_2} + \frac{\hat{\omega}_{\{1,3\}}}{\lambda_1 + \lambda_3} \\ \frac{\hat{\omega}_{\{1,2\}}}{\lambda_1 + \lambda_2} + \frac{\hat{\omega}_{\{2,3\}}}{\lambda_2 + \lambda_3} \\ \frac{\hat{\omega}_{\{1,3\}}}{\lambda_1 + \lambda_3} + \frac{\hat{\omega}_{\{2,3\}}}{\lambda_2 + \lambda_3} \end{pmatrix} - n\bar{t} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \quad (7.42)$$

This has a closed form solution given by

$$\hat{\lambda}_{\mathbf{n}} = \frac{1}{n\bar{t}} \begin{pmatrix} \hat{\omega}_{\{1,2\}} + \hat{\omega}_{\{1,3\}} - \hat{\omega}_{\{2,3\}} \\ \hat{\omega}_{\{1,2\}} - \hat{\omega}_{\{1,3\}} + \hat{\omega}_{\{2,3\}} \\ -\hat{\omega}_{\{1,2\}} + \hat{\omega}_{\{1,3\}} + \hat{\omega}_{\{2,3\}} \end{pmatrix}. \quad (7.43)$$

The information matrix is given by

$$\mathcal{I}(\boldsymbol{\lambda}^*) = \frac{1}{2(\lambda_1^* + \lambda_2^* + \lambda_3^*)} \begin{bmatrix} \frac{1}{\lambda_1^* + \lambda_2^*} + \frac{1}{\lambda_1^* + \lambda_3^*} & \frac{1}{\lambda_1^* + \lambda_2^*} & \frac{1}{\lambda_1^* + \lambda_3^*} \\ \frac{1}{\lambda_1^* + \lambda_2^*} & \frac{1}{\lambda_1^* + \lambda_2^*} + \frac{1}{\lambda_2^* + \lambda_3^*} & \frac{1}{\lambda_2^* + \lambda_3^*} \\ \frac{1}{\lambda_1^* + \lambda_3^*} & \frac{1}{\lambda_2^* + \lambda_3^*} & \frac{1}{\lambda_1^* + \lambda_3^*} + \frac{1}{\lambda_2^* + \lambda_3^*} \end{bmatrix} \quad (7.44)$$

To derive the variance-covariance of the sampling distribution of $\hat{\boldsymbol{\lambda}}_{\mathbf{n}}$ for a sample of n masked system failure times, denoted by $\boldsymbol{\Lambda}_{\mathbf{n}}$, we take the inverse of $n \cdot \mathcal{I}(\boldsymbol{\lambda}^*)$ resulting in

$$\text{Var}_{\boldsymbol{\lambda}^*}[\boldsymbol{\Lambda}_{\mathbf{n}}] = \frac{\lambda_1^* + \lambda_2^* + \lambda_3^*}{n} \begin{bmatrix} \lambda_1^* + \lambda_2^* + \lambda_3^* & -\lambda_3^* & -\lambda_2^* \\ -\lambda_3^* & \lambda_1^* + \lambda_2^* + \lambda_3^* & -\lambda_1^* \\ -\lambda_2^* & -\lambda_1^* & \lambda_1^* + \lambda_2^* + \lambda_3^* \end{bmatrix}. \quad (7.45)$$

By the asymptotic unbiasedness of $\hat{\boldsymbol{\lambda}}_{\mathbf{n}}$, the asymptotic mean squared error is given by the trace of the variance-covariance matrix,

$$\text{MSE}(\boldsymbol{\Lambda}_{\mathbf{n}}) = \frac{3(\lambda_1^* + \lambda_2^* + \lambda_3^*)^2}{n}. \quad (7.46)$$

According to theorem 5.1, for a sufficiently large sample size n ,

$$\boldsymbol{\Lambda}_{\mathbf{n}} \sim \text{MVN}\left(\boldsymbol{\lambda}^*, \frac{1}{n} \mathcal{I}^{-1}(\boldsymbol{\lambda}^*)\right). \quad (7.47)$$

Consequently, each time we observe a particular $\boldsymbol{\Lambda}_{\mathbf{n}} = \hat{\boldsymbol{\lambda}}_{\mathbf{n}}$, we draw a random vector from the multivariate normal distribution given by eq. (7.47). Thus, an independent asymptotic $(1-\alpha) \cdot 100\%$ -confidence interval for λ_j^* is given by

$$\hat{\lambda}_j \pm \frac{z_{1-\alpha/2}}{\sqrt{n}} (\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3). \quad (7.48)$$

Component sets of size one

If we condition on samples in which each observation consists of one candidate, $W = |C| = 1$, the likelihood function is given by

$$\mathcal{L}(\boldsymbol{\lambda} | \bar{t}, \hat{\boldsymbol{\omega}}) = \lambda_1^{\hat{\omega}_{\{1\}}} \lambda_2^{\hat{\omega}_{\{2\}}} \lambda_3^{\hat{\omega}_{\{3\}}} \exp\left(-n\bar{t}(\lambda_1 + \lambda_2 + \lambda_3)\right) \quad (7.49)$$

and the log-likelihood function is given by

$$\ell(\boldsymbol{\lambda} | \bar{t}, \hat{\boldsymbol{\omega}}) = \hat{\omega}_{\{1\}} \ln \lambda_1 + \hat{\omega}_{\{2\}} \ln \lambda_2 + \hat{\omega}_{\{3\}} \ln \lambda_3 - n\bar{t}(\lambda_1 + \lambda_2 + \lambda_3). \quad (7.50)$$

The maximum likelihood estimator is given by

$$\hat{\boldsymbol{\lambda}}_{\mathbf{n}} = \frac{1}{n\bar{t}} \begin{pmatrix} \hat{\omega}_{\{1\}} \\ \hat{\omega}_{\{2\}} \\ \hat{\omega}_{\{3\}} \end{pmatrix}, \quad (7.51)$$

and the information matrix is given by

$$\mathcal{I}(\boldsymbol{\lambda}^*) = \frac{1}{\lambda_1^* + \lambda_2^* + \lambda_3^*} \begin{bmatrix} \frac{1}{\lambda_1^*} & 0 & 0 \\ 0 & \frac{1}{\lambda_2^*} & 0 \\ 0 & 0 & \frac{1}{\lambda_3^*} \end{bmatrix}, \quad (7.52)$$

and thus the variance-covariance matrix is given by

$$\Sigma_{\mathbf{n}}(\boldsymbol{\lambda}^*) = \frac{\lambda_1^* + \lambda_2^* + \lambda_3^*}{n} \begin{bmatrix} \lambda_1^* & 0 & 0 \\ 0 & \lambda_2^* & 0 \\ 0 & 0 & \lambda_3^* \end{bmatrix}. \quad (7.53)$$

The asymptotic mean squared error is given by the trace of the variance-covariance matrix,

$$\text{MSE}(\mathbf{\Lambda}_{\mathbf{n}}) = \frac{(\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3)^2}{n}, \quad (7.54)$$

which has three times less mean squared error than when conditioning on $|C| = 2$. That is, an $\mathbf{M}_{\mathbf{n}}$ sample in which each observation consists of one candidate has more information about $\boldsymbol{\lambda}^*$ than a sample in which each observation consists of two candidates.

The independent asymptotic $(1 - \alpha) \cdot 100\%$ -confidence interval for λ_j^* is given by

$$\hat{\lambda}_j \pm z_{1-\alpha/2} \sqrt{\frac{\hat{\lambda}_j (\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3)}{n}}. \quad (7.55)$$

By comparison, the confidence interval when conditioning on $|C| = 2$ is

$$\sqrt{\frac{\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3}{\hat{\lambda}_j}} \quad (7.56)$$

as wide.

Degenerate: component sets of size three

The degenerate case is all components are candidates, $W = |C| = 3$. There is no maximum likelihood estimator $\hat{\boldsymbol{\lambda}}_{\mathbf{n}}$. Rather, solving the equation results in the underspecified system given by

$$\sum_{j=1}^3 \hat{\lambda}_j - \frac{1}{\bar{t}} = 0, \quad (7.57)$$

where $\hat{\lambda}_j > 0$ for $j = 1, 2, 3$.

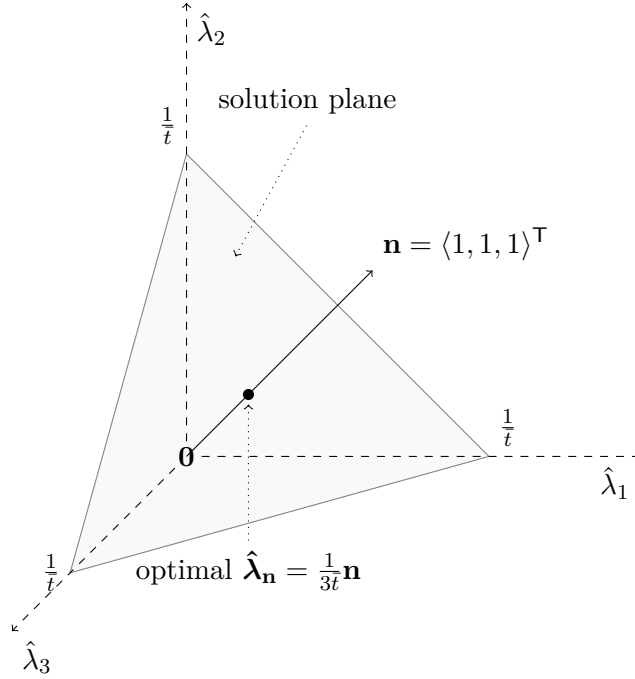
The solution set of this underspecified system is the equation of a plane bounded in the positive region as depicted in fig. 3. The plane has a normal vector given by

$$\mathbf{n} = \langle 1, 1, 1 \rangle^T, \quad (7.58)$$

which is independent of the statistic $\bar{t}$. Across multiple samples, the statistic $\bar{t}$ varies but the normal vector of the plane does not. Thus, the sampling distribution of the solution set will be a density over planes bounded in the positive region with a normal vector given by eq. (7.58).

As n goes to infinity, the probability that $\boldsymbol{\lambda}^*$ intersects the plane goes to 1. This information may not be very useful, especially if the system lifetime S has a small expected value (the area of the bounded plane is $1/\bar{t}^2$), but if this is the only information available, what is the best course of action? An unbiased estimator that asymptotically converges to $\boldsymbol{\lambda}^*$ is not possible, but we can minimize some other loss function. If we assume $\boldsymbol{\lambda}^*$ takes on any supported value with equal likelihood, then the estimate that minimizes the expected Euclidean distance from the projection of $\boldsymbol{\lambda}^*$ onto the plane of the solution set is given by

$$\tilde{\boldsymbol{\lambda}}_{\mathbf{n}} = \frac{1}{3\bar{t}} \mathbf{n}. \quad (7.59)$$

Figure 3: $\hat{\lambda}_{\mathbf{n}}$ solutions given three candidates


7.5.2 Rate of convergence to the asymptotic sampling distribution

In this section, we evaluate how rapidly the sampling distribution of $\hat{\theta}(n|w)$ converges to the asymptotic distribution as given by ??, a multivariate normal with a mean θ^* and a variance-covariance matrix $\frac{1}{n} \mathcal{I}^{-1}(\theta^*)$.

By eq. (5.16), the asymptotic mean squared error of the maximum likelihood estimator $\hat{\theta}(n|w)$ is given by the trace of the variance-covariance matrix, $\text{MSE} = \frac{1}{n} \text{tr} \left(\mathcal{I}^{-1}(\theta^*) \right)$. An estimator of the mean squared error from a sample of r maximum likelihood estimates is given by

$$\hat{\text{MSE}} \left(\hat{\theta}^{(1)}(n|w), \dots, \hat{\theta}^{(r)}(n|w) \right) = \frac{1}{r} \sum_{i=1}^r \left(\hat{\theta}^{(i)}(n|w) - \theta^* \right)^{\top} \left(\hat{\theta}^{(i)}(n|w) - \theta^* \right). \quad (7.60)$$

By generating r masked system failure time of samples of size n , denoted by $\mathbf{M}(n, 1), \dots, \mathbf{M}(r, 1)$, and finding the corresponding maximum likelihood estimates, denoted by $\hat{\theta}^{(1)}(n|w), \dots, \hat{\theta}^{(r)}(n|w)$, we may estimate the mean squared error and compare the result against the asymptotic mean squared error. When we perform these steps for masked system failure time samples of size $n = 1, \dots, N$, we may plot the rate of convergence.

The absolute difference between the mean squared error of the asymptotic covariance matrix and the “true” covariance matrix is given by

$$|\text{MSE} - \hat{\text{MSE}}|, \quad (7.61)$$

*** Let’s use a profile likelihood to look at how the likelihood function changes with respect to a parameter component. Say, for instance, the other components in the parameter are nuisance parameters, and we are really just interested in the j^{th} component. ***

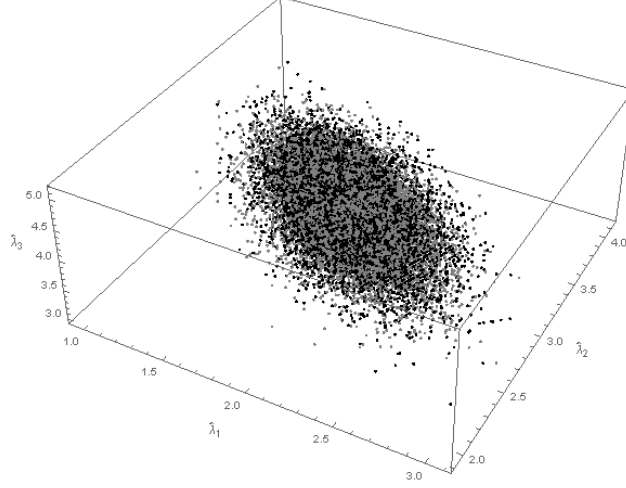


Figure 4: Theoretical MLE data points (gray) vs actual MLE data points (black)

Let $\boldsymbol{\lambda}^* = \langle 2, 3, 4 \rangle^\top$ and let the sample of masked system failure times be of size $n = 1000$. Theoretically, the MLE $\hat{\boldsymbol{\lambda}}_{1000}$ is approximately distributed as

$$\hat{\boldsymbol{\lambda}}_{1000} \sim \text{MVN}\left(\langle 2, 3, 4 \rangle^\top, \boldsymbol{\Sigma}_{1000}\right). \quad (7.62)$$

where

$$\boldsymbol{\Sigma}_{1000} = \begin{pmatrix} 0.081 & -0.036 & -0.027 \\ -0.036 & 0.081 & -0.018 \\ -0.027 & -0.018 & 0.081 \end{pmatrix}. \quad (7.63)$$

We generated two different sets of data points, one from the theoretical distribution described in eq. (7.62) and the other from the maximum likelihood method applied to $\mathbf{M}_{1000}$ samples. Refer to fig. 4 for a visualization of the data points. Visually, they have a similar distribution of data points, indicating that the asymptotic distribution is a reasonable approximation of the sampling distribution of the MLE $\hat{\boldsymbol{\lambda}}_{1000}$.

A sample variance-covariance matrix of $r = 10000$ maximum likelihood estimates was computed to be

$$\widehat{\boldsymbol{\Sigma}}_{1000} = \begin{pmatrix} 0.081 & -0.037 & -0.027 \\ -0.037 & 0.082 & -0.018 \\ -0.027 & -0.018 & 0.081 \end{pmatrix}, \quad (7.64)$$

which is approximately equivalent to the theoretical asymptotic distribution given by eq. (7.62).

To derive an accurate estimate of the sampling distribution of $\hat{\boldsymbol{\lambda}}_{\mathbf{n}}$ for small n , as opposed to the asymptotic sampling distribution for large n , we generate r masked system failure time samples

We define the error between two matrices $\mathbf{A}$ and $\mathbf{B}$ to be given by

$$\|\mathbf{A} - \mathbf{B}\|_F, \quad (7.65)$$

where $\|\mathbf{C}\|_F$ is the Frobenius norm of matrix $\mathbf{C}$,

$$\|\mathbf{C}\|_F = \sqrt{\text{tr}(\mathbf{C}\mathbf{C}^\top)} \quad (7.66)$$

$$= \sqrt{\text{vec}(\mathbf{A})^\top \text{vec}(\mathbf{A})}. \quad (7.67)$$

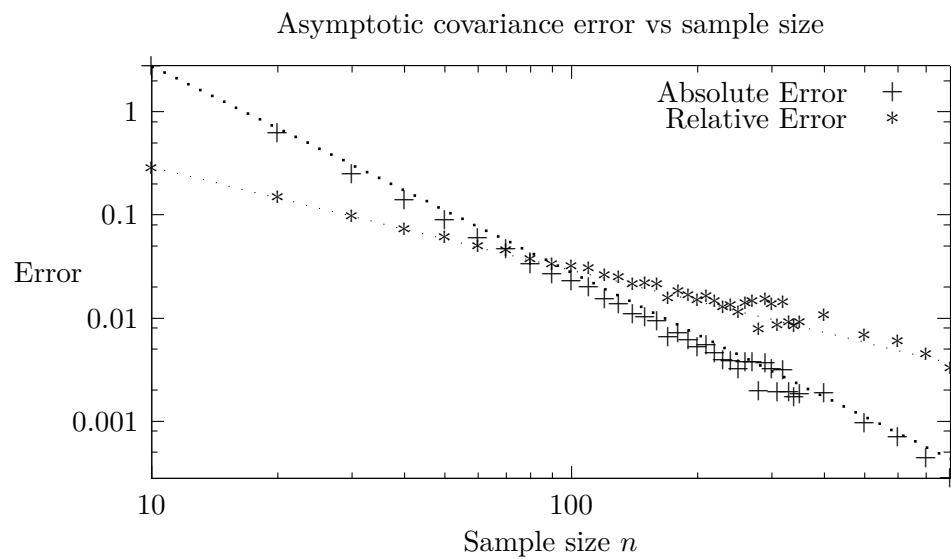
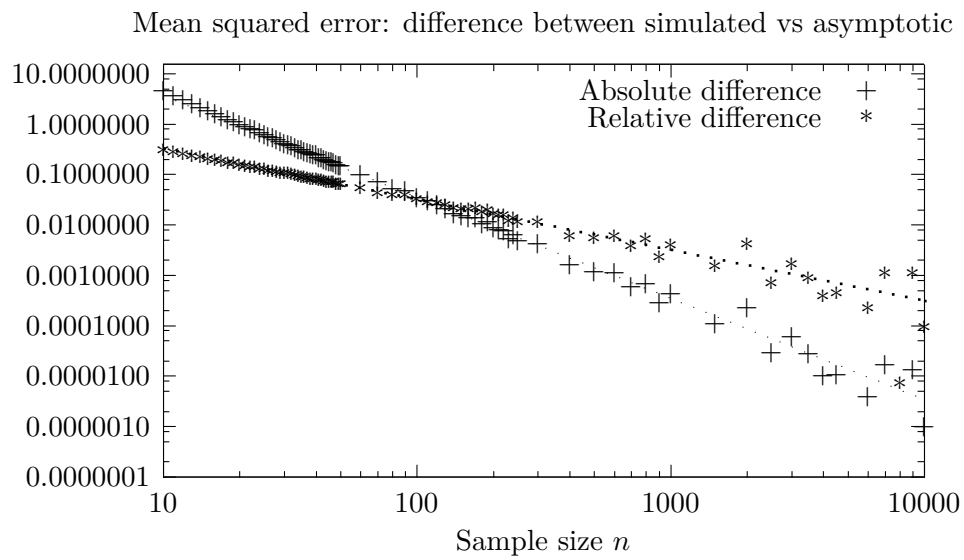
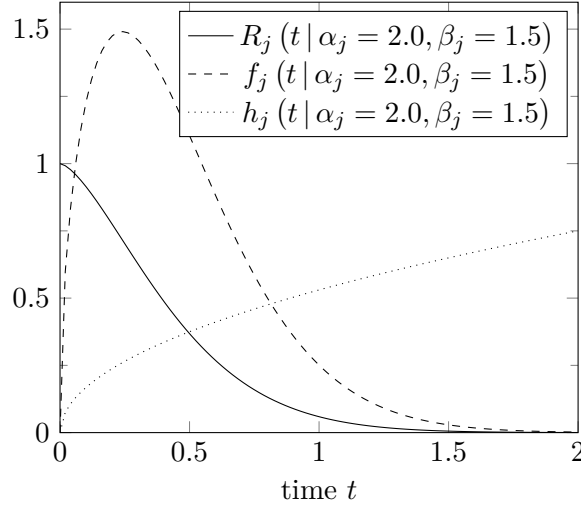


Figure 5: Weibull



In section 7.5.2, we log-log plot the error between a simulated estimate of the true covariance matrix and the asymptotic covariance matrix with respect to sample size. The errors approximately follow a straight line, suggesting a power law error. We fit a power function of the form an^b to the differences with respect to sample size n and found the best fit for the relative difference to be approximately proportional to $\frac{1}{n}$ and the best fit for the absolute difference to be approximately proportional to $\frac{1}{n^2}$.

8 Case study: Weibull

In our second case study, we analyze a series system in which the components have Weibull distributed lifetimes. Consider a component j with a lifetime $T_j \sim \text{WEI}(\alpha_j^*, \beta_j^*)$. The random variable T_j has reliability, density, and failure rate functions given by

$$R_j(t | \alpha_j, \beta_j) = \exp \left[- \left(\frac{t}{\alpha_j} \right)^{\beta_j} \right], \quad (8.1)$$

$$f_j(t | \alpha_j, \beta_j) = \frac{\beta_j}{\alpha_j} \left(\frac{t}{\alpha_j} \right)^{\beta_j-1} \exp \left[- \left(\frac{t}{\alpha_j} \right)^{\beta_j} \right], \quad (8.2)$$

$$h_j(t | \alpha_j, \beta_j) = \frac{\beta_j}{\alpha_j} \left(\frac{t}{\alpha_j} \right)^{\beta_j-1}, \quad (8.3)$$

where $t > 0$, $\alpha_j > 0$ is the scale parameter, and $\beta_j > 0$ is the shape parameter. See fig. 5 for a visualization of these functions. By eq. (8.3), a component j has a failure rate that increases with respect to time t (e.g., wears out) if $\beta_j > 1$, decreases (e.g., burn-in period) if $0 < \beta_j < 1$, and remains constant if $\beta_j = 1$.¹

¹ $T_j \sim \text{WEI}(\alpha_j, \beta_j = 1) \iff T_j \sim \text{EXP}(1/\alpha_j)$.

The lifetime of the series system composed of m components with Weibull distributed lifetimes, $T_j \sim \text{WEI}(\alpha_j^*, \beta_j^*)$ for $j = 1, \dots, m$, has a true parameter value given by

$$\Theta^* = \begin{pmatrix} \alpha_1^* & \beta_1^* \\ \vdots & \vdots \\ \alpha_m^* & \beta_m^* \end{pmatrix}. \quad (8.4)$$

where component j has a true parameter value indexed by row j of Θ^* , denoted by Θ_j^* . The system's reliability is given by

$$R_S(t | \Theta^*) = \exp \left[- \sum_{j=1}^m \left(\frac{t}{\alpha_j} \right)^{\beta_j} \right]. \quad (8.5)$$

Proof. By theorem 3.1,

$$R_S(t | \Theta^*) = \prod_{j=1}^m R_j(t | \Theta_j^*). \quad (a)$$

Plugging in the component reliability functions as given by eq. (8.1) results in

$$R_S(t | \Theta^*) = \prod_{j=1}^m \exp \left[- \left(\frac{t}{\alpha_j} \right)^{\beta_j} \right] = \exp \left[- \sum_{j=1}^m \left(\frac{t}{\alpha_j} \right)^{\beta_j} \right]. \quad (b)$$

□

The system's failure rate is given by

$$h_S(t | \Theta^*) = \sum_{j=1}^m \frac{\beta_j}{\alpha_j} \left(\frac{t}{\alpha_j} \right)^{\beta_j-1}, \quad (8.6)$$

whose proof follows from theorem 3.1. The Weibull series system can have failure rates depicted by fig. 6.

By definition, the system's density is given by

$$f_S(t | \Theta^*) = h_S(t | \Theta^*) R_S(t | \Theta^*). \quad (8.7)$$

where R_S and h_S are given respectively by Equations (8.1) and (8.6).

By ??, the conditional probability that component k is the cause of a system failure at time t is given by

$$p_{K|S}(k | t, \Theta^*) = \frac{h_k(t | \Theta_k^*)}{h_S(t | \Theta^*)}, \quad (8.8)$$

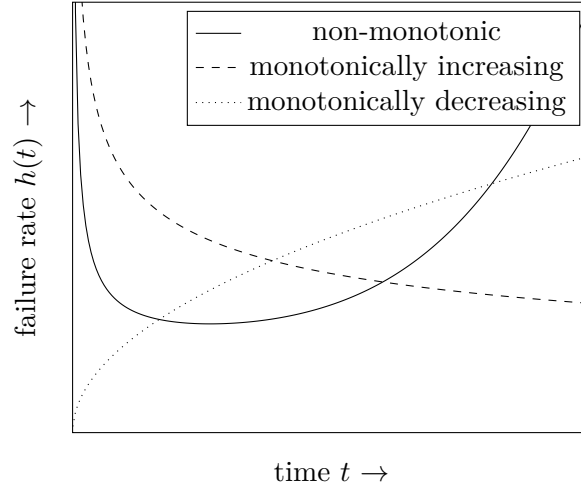
where h_k and h_S are respectively given by eqs. (8.3) and (8.7).

By definition, the joint density of K and S is given by

$$f_{K,S}(k, t | \Theta^*) = p_{K|S}(k | t, \Theta^*) f_S(t | \Theta^*), \quad (8.9)$$

where conditional probability mass and the marginal probability are respectively given by eq. (8.8) and ??.

Figure 6: Failure rate monotonicity



By ??, the joint density of C and S is given by

$$f_{C,S}(\mathcal{C}, t | \Theta^*) = \binom{m-1}{w-1}^{-1} \sum_{k \in \mathcal{C}} f_{K,S}(k, t | \Theta^*). \quad (8.10)$$

For subsequent material, we prefer to work with vectors rather than matrices. By definition 4.1, $\theta \equiv \text{vec}(\Theta) = \langle \alpha_1, \beta_1, \alpha_2, \beta_2, \dots, \alpha_m, \beta_m \rangle$. The joint likelihood $\mathcal{L}$ for a masked system failure sample $\mathbf{M}_n$ is given by

$$\mathcal{L}(\theta | \mathbf{M}_n) = \prod_{i=1}^n f_{C,S}(\mathcal{C}_i, t_i | \theta) \quad (8.11)$$

and the log-likelihood is given by

$$\ell(\theta | \mathbf{M}_n) = \sum_{i=1}^n \ln f_{C,S}(\mathcal{C}_i, t_i | \theta). \quad (8.12)$$

The score, the gradient of the log-likelihood ℓ , is given by

$$s(\theta) = \nabla_{\theta} \ell(\theta | \mathbf{M}_n) = \left\langle \frac{\partial \ell}{\partial \alpha_1}, \frac{\partial \ell}{\partial \beta_1}, \dots, \frac{\partial \ell}{\partial \alpha_m}, \frac{\partial \ell}{\partial \beta_m} \right\rangle. \quad (8.13)$$

The Weibull series system has no closed form solution so instead we use the observed information matrix $\mathcal{J}_n$, which computes the covariance of $\hat{\Theta}_n$ conditioned on an observed sample $\mathbf{M}_n = \mathbf{m}_n$,

$$\mathcal{J}_n(\Theta^* | \mathbf{m}_n) = -\mathcal{H}(\ell) \quad (8.14)$$

Bibliography

- [1] Peter Bickel and Kjell Doksum. *Mathematical Statistics*, volume 1, chapter 2, page 117. Prentice Hall, 2 edition, 2000.

A Alternative proofs

A.1 Equation 3.6

Equation (3.6) on page 10 asserts that

$$f_{K,S}(k, t | \Theta^*) = f_k(t | \Theta_k^*) \prod_{\substack{j=1 \\ j \neq k}}^m R_j(t | \Theta_j^*).$$

Proof. First, note that $f_{K,S}(k, t | \Theta^*)$ is a proper density since

$$\int_0^\infty \sum_{j=1}^m f_{K,S}(j, t | \Theta^*) dt = \int_0^\infty f_S(t | \Theta^*) dt = 1. \quad (a)$$

Consider a 3-out-of-3 system. By assumption 2, T_1 , T_2 , and T_3 are mutually independent. Thus, the joint density that $T_1 = t_1$, $T_2 = t_2$, and $T_3 = t_3$ is given by

$$f_{T_1, T_2, T_3}(t_1, t_2, t_3 | \Theta^*) = \prod_{p=1}^3 f_p(t_p | \Theta_p^*). \quad (b)$$

Component 1 causes a system failure during the interval $(0, t)$, $t > 0$, if component 1 fails during the given interval and components 2 and 3 survive longer than component 1. That is, $0 < T_1 < t$, $T_1 < T_2$, and $T_1 < T_3$.

Let $p(t)$ denote the probability that component 1 causes a system failure during the interval $(0, t)$,

$$p(t) = P[0 < T_1 < t \cap T_1 < T_2 \cap T_1 < T_3]. \quad (c)$$

This probability is given by

$$p(t) = \int_{t_1=0}^{t_1=t} \int_{t_2=t_1}^{t_2=\infty} \int_{t_3=t_1}^{t_3=\infty} \prod_{p=1}^3 f_p(t_p | \Theta_p^*) dt_3 dt_2 dt_1. \quad (d)$$

Performing the integration over t_3 results in

$$p(t) = \int_{t_1=0}^{t_1=t} \int_{t_2=t_1}^{t_2=\infty} \prod_{p=1}^2 f_p(t_p | \Theta_p^*) R_3(t_1 | \Theta_3^*) dt_2 dt_1. \quad (e)$$

Performing the integration over t_2 results in

$$p(t) = \int_{t_1=0}^{t_1=t} f_1(t_1 | \Theta_1^*) \prod_{p=2}^3 R_p(t_1 | \Theta_p^*) dt_1. \quad (f)$$

Taking the derivative of $p(t)$ with respect to t produces the density of probability near t . By the Second Fundamental Theorem of Calculus,

$$\frac{dp}{dt} = f_1(t | \Theta_1^*) \prod_{p=2}^3 R_p(t | \Theta_p^*). \quad (g)$$

The probability density $\frac{dp}{dt}$ is equivalent to the joint density that $K = 1$ and $S = t$ as given by theorem 3.4, i.e., $\frac{dp}{dt} = f_{K,S}(1, t | \Theta^*)$. Generalizing from this completes the proof. $\square$

A.2 Equation ??

By ??, S and W are independent and the marginal distribution of W is independent of θ^* , thus the likelihood of observing a particular realization, $\mathbf{M}_n = \mathbf{m}_n$, is given by

$$\begin{aligned} f_{\mathbf{M}_n}(\mathbf{m}_n | \theta^*) &= \prod_{i=1}^n f_{C,S|W}(\mathcal{C}_i, t_i | |\mathcal{C}_i|, \theta^*) p_W(|\mathcal{C}_i|) \\ &= \left[\prod_{i=1}^n f_{C,S|W}(\mathcal{C}_i, t_i | |\mathcal{C}_i|, \theta^*) \right] \left[\prod_{i=1}^n p_W(|\mathcal{C}_i|) \right]. \end{aligned} \quad (\text{A.1})$$

If we fix $\mathbf{m}_n$ and allow the parameter θ to change, we have the likelihood function

$$\mathcal{L}(\theta | \mathbf{m}_n) = c \prod_{i=1}^n f_{C,S|W}(\mathcal{C}_i, t_i | |\mathcal{C}_i|, \theta), \quad (\text{A.2})$$

where the product over $p_W(\cdot)$ is constant with respect to θ and has been relabeled as c . The log-likelihood with respect to θ then is given by

$$\begin{aligned} \ell(\theta | \mathbf{m}_n) &= \ln c + \sum_{i=1}^n \ln f_{C,S|W}(\mathcal{C}_i, t_i | |\mathcal{C}_i|, \theta) \\ &= c' + \sum_{w=1}^{m-1} \left[\sum_{i \in \mathbb{A}(w)} \ln f_{C,S|W}(\mathcal{C}_i, t_i | w, \theta) \right], \end{aligned} \quad (\text{A.3})$$

where $\mathbb{A}(w) = \{i \in \{1, \dots, n\} : |\mathcal{C}_i| = w\}$. Let $\mathbf{m}_{n_w} = \{(\mathcal{C}_i, t_i) : i \in \mathbb{A}(w)\}$. The part in brackets is the log-likelihood $\ell(\theta | \mathbf{m}_{n_w})$. Performing this substitution yields

$$\ell(\theta | \mathbf{m}_n) = \ln c + \sum_{w=1}^{m-1} \ell(\theta | \mathbf{m}_{n_w}). \quad (\text{A.4})$$

The score function is given by

$$s(\theta | \mathbf{m}_n) = \sum_{w=1}^{m-1} s(\theta | w, \mathbf{m}_{n_w}), \quad (\text{A.5})$$

The information matrix may be computed by taking the negative of the expectation of the Jacobian of the score over the true joint distribution of S, C, and W, giving

$$\mathcal{I}(\theta^*) = -\mathbb{E}_{\theta^*} \left[\mathcal{J} \left(s(\theta | \mathbf{M}_1) \right) \Big|_{\theta=\theta^*} \right]. \quad (\text{A.6})$$

Substituting eq. (A.5) with its definition gives

$$\mathcal{I}(\theta^*) = -\mathbb{E}_{\theta^*} \left[\mathcal{J} \left(\sum_{w=1}^{m-1} s(\theta | w, \mathbf{M}(w, 1)) \right) \Big|_{\theta=\theta^*} \right]. \quad (\text{A.7})$$

The Jacobian is a linear operator so it can be moved inside the summation, giving

$$\mathcal{I}(\boldsymbol{\theta}^*) = -\mathbb{E}_{\boldsymbol{\theta}^*} \left[\sum_{w=1}^{m-1} \mathcal{J}(\mathbf{s}(\boldsymbol{\theta} | w, \mathbf{M}(w, 1))) \right]_{\boldsymbol{\theta}=\boldsymbol{\theta}^*}. \quad (\text{A.8})$$

The Jacobian of $\mathbf{s}$ is the Hessian of ℓ . Performing this substitution gives

$$\mathcal{I}(\boldsymbol{\theta}^*) = -\mathbb{E}_{\boldsymbol{\theta}^*} \left[\sum_{w=1}^{m-1} \mathcal{H}(\ell(\boldsymbol{\theta} | w, \mathbf{M}(w, 1))) \right]_{\boldsymbol{\theta}=\boldsymbol{\theta}^*}. \quad (\text{A.9})$$

The expectation is a linear operator so it may be moved inside the summation, giving

$$\mathcal{I}(\boldsymbol{\theta}^*) = -\sum_{w=1}^{m-1} \mathbb{E}_{\boldsymbol{\theta}^*} \left[\mathcal{H}(\ell(\boldsymbol{\theta} | w, \mathbf{M}(w, 1))) \right]_{\boldsymbol{\theta}=\boldsymbol{\theta}^*}. \quad (\text{A.10})$$

The expectation sums over the probability mass function $\mathbf{W} \sim p_{\mathbf{W}}(\cdot)$, giving

$$\mathcal{I}(\boldsymbol{\theta}^*) = -\sum_{w=1}^{m-1} p_{\mathbf{W}}(w) \mathbb{E}_{\boldsymbol{\theta}^*} \left[\mathcal{H}(\ell(\boldsymbol{\theta} | w, \mathbf{M}(w, 1))) \right], \quad (\text{A.11})$$

where the expectation is now over the true marginal joint distribution of $\mathbf{C}$ and $\mathbf{S}$ given $\mathbf{W} = w$. The expectation of the Hessian of ℓ is equivalent to the negative of $\mathcal{I}(\boldsymbol{\theta}^* | w, \mathbf{M}(w, 1))$. Performing this substitution gives

$$\mathcal{I}(\boldsymbol{\theta}^*) = \sum_{w=1}^{m-1} p_{\mathbf{W}}(w) \mathcal{I}(\boldsymbol{\theta}^* | w). \quad (\text{A.12})$$

□

B Numerical solutions to the MLE

The function $\ell(\boldsymbol{\theta} | w, \mathbf{m}_{\mathbf{n}})$ is the log-likelihood with respect to $\boldsymbol{\theta} \in \Omega$ where $\Omega \subset \mathbb{R}^{m \cdot q}$. This function has a surface in an $(m \cdot q + 1)$ dimensional space, where a particular point on this surface represents the log-likelihood of observing $\mathbf{m}_{\mathbf{n}}$ with respect to $\boldsymbol{\theta}$. By definition 5.3, $\hat{\boldsymbol{\theta}}(n | w)$ is the point on this surface that is at a maximum,

$$\hat{\boldsymbol{\theta}}(n | w) = \arg \max_{\boldsymbol{\theta} \in \Omega} \ell(\boldsymbol{\theta} | w, \mathbf{m}_{\mathbf{n}}). \quad (\text{5.9 revisited})$$

Generally there is no closed-form solution that solves $\hat{\boldsymbol{\theta}}(n | w)$, in which case iterative search methods may be used to numerically approximate a solution.

The general version of iterative search that numerically approximates a solution to eq. (5.10), subject to the constraint $\boldsymbol{\theta}^* \in \Omega$, is shown in algorithm 5. Since iterative search is a local search method, it may fail to converge to a global maximum.

Assume parameter space Ω is convex. Then, the function $\text{project}(\boldsymbol{\theta}, \Omega)$ in algorithm 5 projects any point $\boldsymbol{\theta}$ to the nearest point in the parameter space Ω as depicted by fig. 7. Thus, the search method is restricted to searching over the feasible parameter space.

Algorithm 5: Iterative maximum likelihood search

Result: an approximate solution to the stationary points of the maximum likelihood equation

Input:

Θ^* , the true parameter index.

Output:

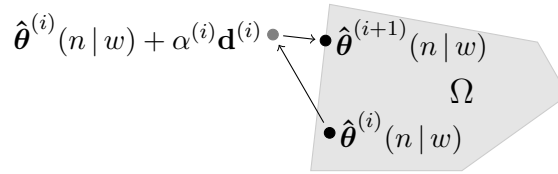
$\hat{\theta}(n|w)$, an approximation of the maximum likelihood estimate.

```

1 Model find_mle( $\mathbf{m}_n$ )
2    $\hat{\theta}^{(0)}(n|w) \leftarrow$  an initial starting point in  $\Omega$ 
3    $i \leftarrow 0$ 
4   while stopping criteria not satisfied do
5      $\hat{\theta}^{(i+1)}(n|w) \leftarrow \text{project} \left( \hat{\theta}^{(i)}(n|w) + \alpha^{(i)} \mathbf{d}^{(i)}, \Omega \right)$ 
6      $i \leftarrow i + 1$ 
7   end
8   return  $\hat{\theta}^{(i)}(n|w)$ 

```

Figure 7: Projection onto convex set Ω .



In algorithm 5, $\mathbf{d}^{(i)}$ is a unit vector denoting a *promising* direction in which to search for a better solution than $\hat{\theta}^{(i)}(n|w)$. The Fisher scoring algorithm is a slight variation of Newton-Raphson² in which

$$\mathbf{d}^{(i)} = \mathcal{I}^{-1} \left(\hat{\theta}^{(i)}(n|w) \right) s \left(\hat{\theta}^{(i)}(n|w) \right). \quad (\text{B.1})$$

The scalar $\alpha^{(i)} > 0$ is the distance to move from point $\hat{\theta}^{(i)}(n|w)$ to generate the next point, $\hat{\theta}^{(i+1)}(n|w)$. Most naturally, the solution to

$$\alpha^{(i)} = \arg \max_{\alpha > 0} \ell(\hat{\theta}^{(i)}(n|w) + \alpha \cdot \mathbf{d}^{(i)}) \quad (\text{B.2})$$

is desired, e.g., using *Golden Section* line search to find an approximate solution. Finally, the iterations are repeated until some *stopping criteria* is satisfied. Most naturally, this is given by

$$\|s(\hat{\theta}^{(i)}(n|w))\| < \epsilon, \quad (\text{B.3})$$

²If the *observed* information matrix is used instead of the *expected* information matrix, the Fisher scoring algorithm is equivalent to Newton-Raphson.

signifying a stationary point has been reached.

C Generic joint distribution

We assume the distribution of W does not carry information about the true parameter index θ^* . By the axioms of probability, the joint distribution of C , S , and W is given by

$$f_{C,S,W}(C, t, w | \theta^*) = p_{C|S,W}(C | t, w, \theta^*) p_{W|S}(w | t) f_S(t | \theta^*). \quad (C.1)$$

Since W carries no information about the true parameter index θ^* , the maximum likelihood estimate $\hat{\theta}_n$ is independent of the distribution of W .

Proof.

$$\begin{aligned} \hat{\theta}_n &= \arg \max_{\theta} \ell(\theta | \mathbf{m}_n) \\ &= \arg \max_{\theta} \left[\sum_{i=1}^n \ln p_{C|S,W}(C_i | t_i, |C_i|, \theta) \right. \\ &\quad \left. + \sum_{i=1}^n \ln f_S(t_i | \theta) + \underbrace{\sum_{i=1}^n \ln p_{W|S}(|C_i| | t_i)}_{\text{constant}} \right], \end{aligned} \quad (a)$$

where the last term is constant with respect to θ and may be dropped from the maximum likelihood equation without changing the point that maximizes the likelihood. $\square$

The sampling distribution of $\hat{\theta}_n$ is a function of $\mathbf{M}_n$ (as opposed to a particular realization), and since $\mathbf{M}_n$ is a function of W (the distribution of the cardinality of candidate sets), the sampling distribution of $\hat{\theta}_n$ is also a function of W .

Given a statistical model, the asymptotic sampling distribution of the maximum likelihood estimator is nearly automatic. The most taxing (and uncertain) part is the *model selection*. Since we may not have much confidence in any particular model of the joint distribution of W and S , the *expected* information matrix is sketchy. In general, the *observed* information matrix is preferred. The sampling distribution of $\hat{\theta}_n$ converges in distribution to a multivariate normal with a mean given by θ^* and a variance-covariance given by the inverse of the *observed* information matrix, written

$$\mathbf{Y}_n \xrightarrow{d} \text{MVN}(\theta^*, \mathcal{J}_n^{-1}(\theta^* | \mathbf{M}_n)). \quad (C.2)$$

If an accurate distribution of $W | S = t$ could be constructed such that it is dependent on θ^* , then the distribution of W in a sample carries extra information about θ^* . A plausible model of W would depend upon the entropy of $K | S = t$.